



OPERATOR'S MANUAL

Version V1.6.2



PLEASE NOTE:

- Read this manual **BEFORE** operating the machine.
- Keep this manual for your reference.
- Go to www.LAIGames.com Click on Support to register your Games and receive future updates.

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SAFETY PRECAUTIONS

The following safety precautions and advisories are used throughout this manual and are defined as follows.

*** WARNING! ***

*Disregarding this text could result in **serious injury**.*

*** CAUTION! ***

*Disregarding this text could result in **damage to the machine**.*

*** NOTE! ***

Is an advisory text to hint at or help you understand more!



BE SURE TO READ THE FOLLOWING



*** WARNING! ***

Always Turn **OFF** Mains AC power and unplug the game, before opening or replacing any parts.

Always When unplugging the game from an electrical outlet, grasp the plug, not the line cord.

Always Connect the Game Cabinet to grounded electrical outlet with a securely connected ground line.

Do Not Install the Game Cabinet outdoors or in areas of high humidity, direct water contact, dust, high heat or extreme cold.

Do Not Install the Game Cabinet in areas that would present any obstacle in case of an emergency, i.e. near fire equipment or emergency exits.

*** CAUTION! ***

Always Use a Digital Multi meter, logic tester or oscilloscope for testing integrated circuit (IC) logic PC boards. The use of a continuity tester is not permitted.

Do Not Connect or disconnect any of the integrated circuit (IC) logic PC boards while the power is **ON**.

Do Not Use any fuse that does not meet the specified rating.

Do Not Subject the game cabinet to extreme temperature variations. Reliability of electrical components deteriorates rapidly over 60 °C.



MACHINE INSTALLATION and INSPECTION

When installing and inspecting “**Stack N Grab**”, be very careful of the following points to ensure that players enjoy the game safely.

- Be sure to turn the power **OFF** before working on the machine.

*** WARNING! ***

Always Turn **OFF** mains power before removing safety covers and refit all safety covers when work is completed.

- Make sure the power cord is not exposed on the surface (floor, ground, etc.) where people walk.
- Check that the rubber glide feet levelers are set evenly on the floor so that the game cabinet is unable to roll and is stable.
- Always make complete connections for the integrated circuit (IC) logic PC Boards and other connectors. Insufficient insertion can damage the electrical components.

*** CAUTION! ***

Before Switching the machine on be sure to check that it has been set on the correct voltage for your area!

Refer To the mains voltage adjustment section of this manual. Machines are normally shipped on 220V AC unless otherwise specified.

- Only qualified personnel should inspect or test the integrated circuit (IC) logic PC Boards.
- If any integrated circuit (IC) logic PC Boards should need servicing. Please contact the nearest **LAI Games** distributor. (Refer to the back page of this manual)



INTRODUCTION

CONGRATULATIONS! You have just purchased “**Stack N Grab**”, it is another great product in the “*Toyzone*” range from **LAI Games**.

With a bright and very attractive cabinet, “**Stack N Grab**” is an exciting innovative new style of game which uses a crane to vend prizes won and which would make a great addition to any location.

We hope you take the time to read this manual and learn about the many features and user-friendly adjustments that can be made to “fine-tune” the game for maximum earning potential.

DESCRIPTION

- The “**Stack N Grab**” is a game where players have to press the “Stack” button to stop the moving blocks on top of the blocks in the previous level. As the blocks get higher the game gets progressively harder. If the player reaches and stacks the top level, they win and the claw is activated. The player then uses the Joystick to maneuver the crane and drop the claw to “Grab” their prize. The claw remains active until a prize has been dropped into the vend area.

PACKAGING

- At delivery, the machine should arrive in good condition. To move the packaged machine for transport or placement, use a forklift and take care not to hit the package or stack heavy objects on top, as this may cause damage to the machine.

CONTENTS

- The “Stack N Grab” cabinet
- Keys: 2 x Front door keys
 2 x back door keys
- Operator’s manual
- IEC Power Cord (In cash box)
- Parts & Accessories (In cash box)



SPECIFICATIONS

DIMENSIONS

- Weight: 240 kg (529lb)
- Height: 2132mm (84")
- Width: 808mm (31-3/4")
- Length: 1150mm (45-1/4")
- Power: Maximum (360 Watts) (220V @ 1.6A) (120V @ 3.3A)

ELECTRIC SUPPLY

The game has the option to operate on a 110V, 120V, 220V or 240V AC 50/60Hz single phase mains electric supply.

The supply must be a three wire grounded supply.

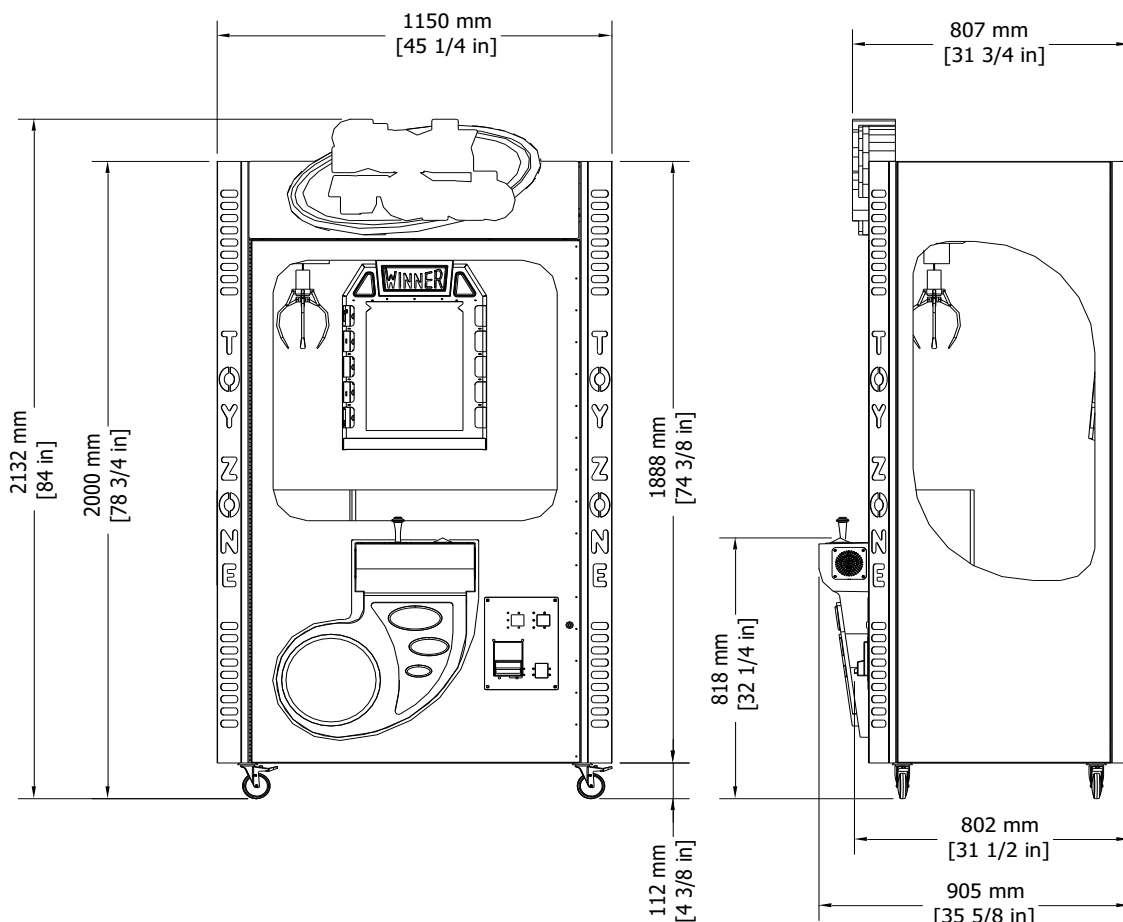
* CAUTION! *

Before Switching the machine on be sure to check that it has been set on the correct voltage for your area!

Please Refer to the mains voltage adjustment section of this manual. Machines are normally shipped on 220V AC unless otherwise specified.

LOCATION REQUIREMENTS

- Ambient temperature: between 5°C and 40°C.
- Ambient humidity: Low
- Ambient U.V. radiation: Very low
- Vibrations level: Low





CASH BOX PARTS

This is the list of parts that can be found inside the cash box when you receive the game,



Spare Rope for the Claw



M2.5 Allen Key



Spare Fuses



Spare M6 Screws



8'' Smaller Claw Kit



ASSEMBLY INSTRUCTIONS

(Step 1)



Unpack the machine
Dispose of the packaging carefully
Keys are in the coin reject
Assembly instructions are inside game
Manual inside the game
Top Header inside the game
Control panel inside the game if disassembled

(Step 2)



Open front door
Take out the Top Header Panel
Take out the control panel if disassembled
Remove the protective cover from the header lighting panel

Keys



Top Header Lighting Panel



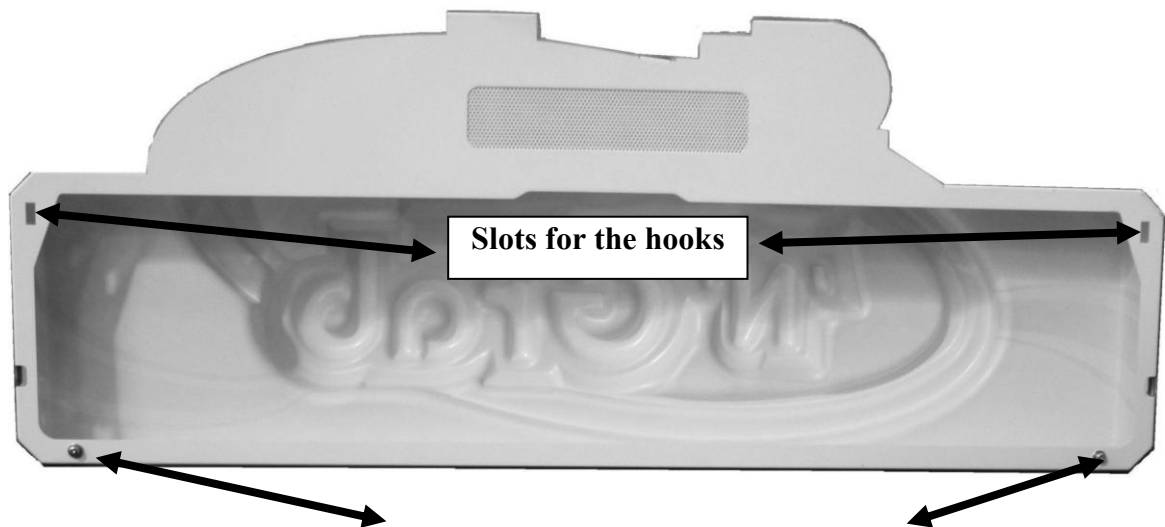
Control Panel



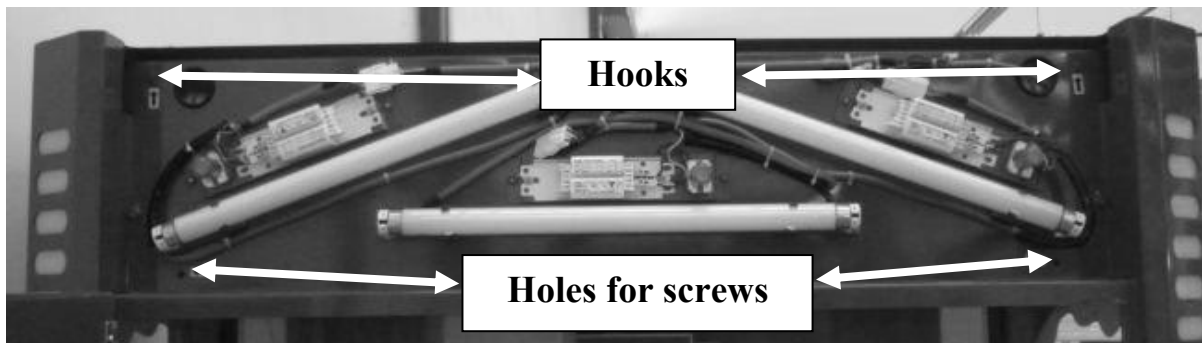
Top Header Panel



(Step 3)



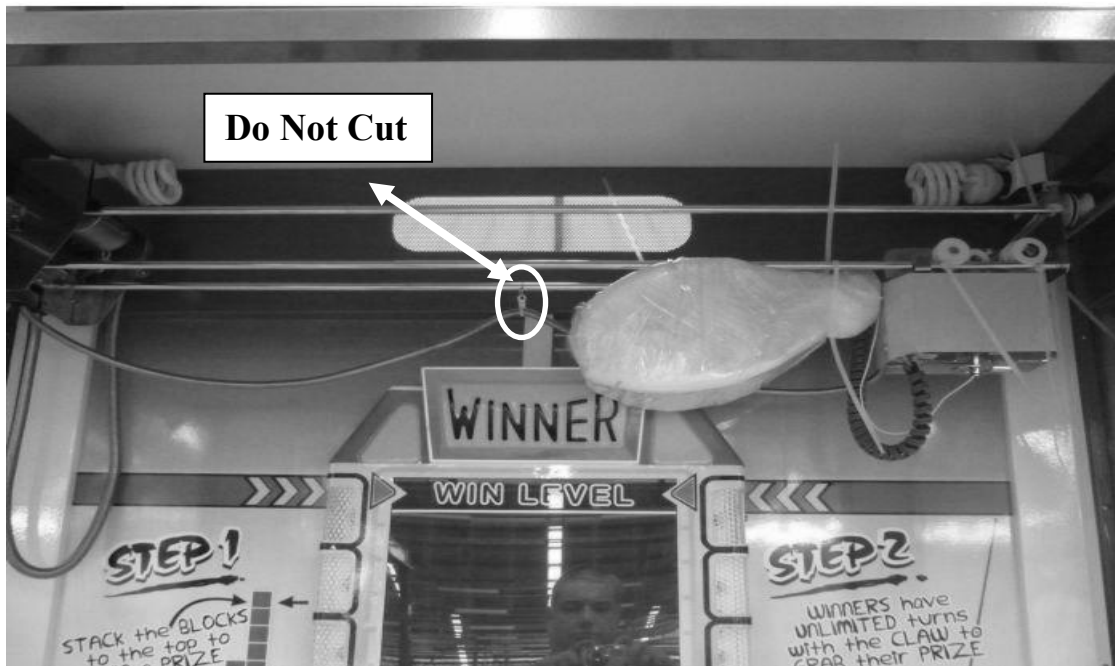
Remove the two screws from the top header panel and keep them safe so you can use them for later to hold the header in place. Look at the top header lighting panel on the game and see two hooks for the header panel to hook onto and from the back, two holes to replace the screws through to secure the header



When Top Header Panel assembled



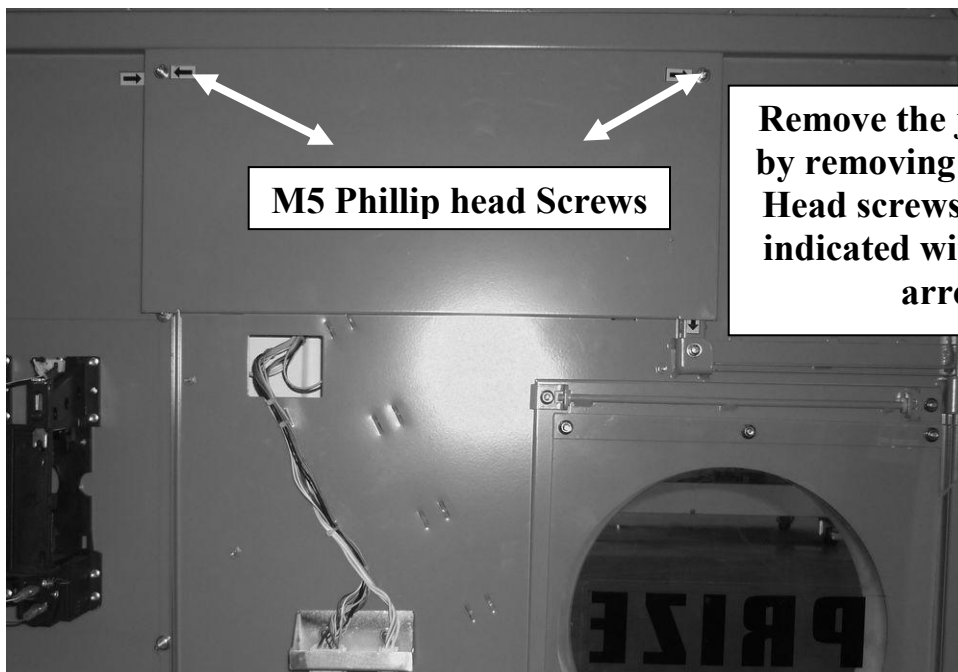
(Step 4)



Remove all the cable ties holding the crane and claw secure for transport.

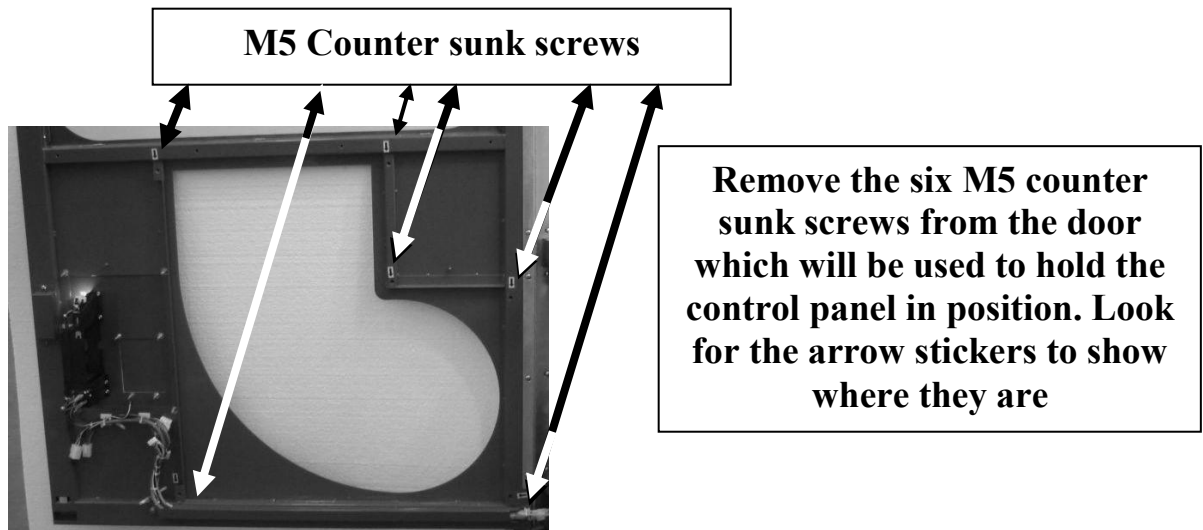
Do not cut the cable tie on the central cable ring support

(Step 5) Only if the control panel has been removed

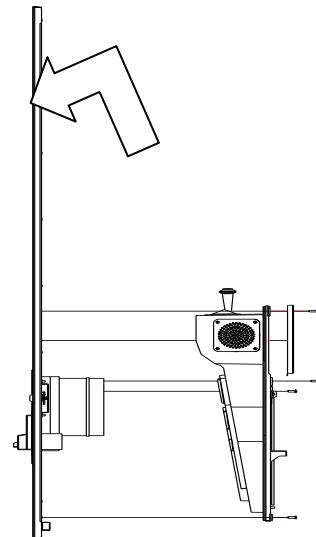




(Step 6)



(Step 7)



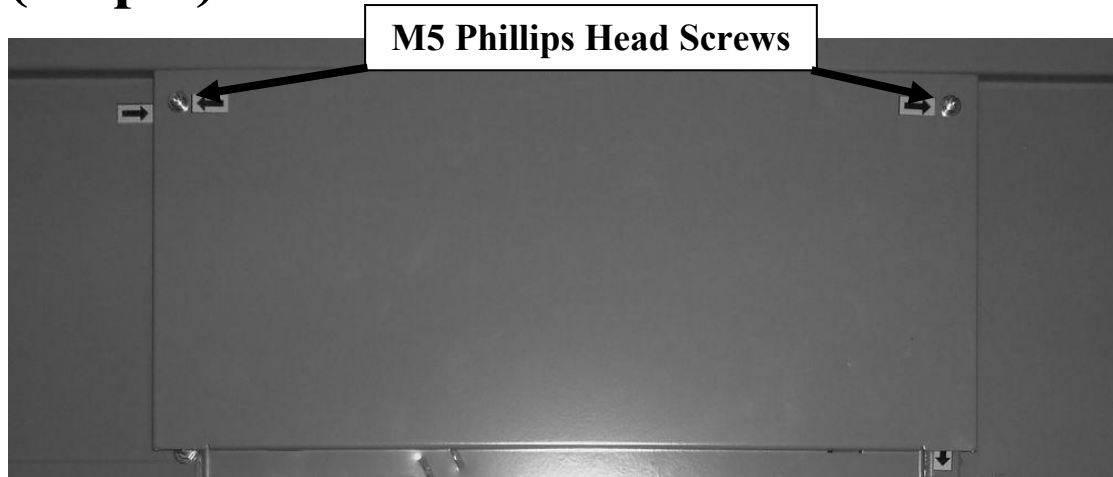
From the inside of the door, fit the control panel by tilting the control panel so the joystick fits through the door.

Hold the control panel in position while you fasten the screws back in to secure the panel

Note: The position from the drawing above



(Step 8)

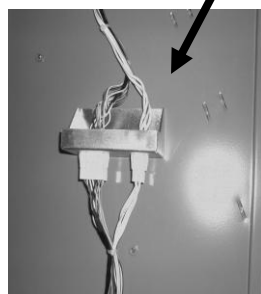


Replace the joystick cover using the two screws you have taken off earlier and replace back to the same position (see above)

(Step 9)



The door should now look like this and will now require the two connectors fastened together



Clip the two connectors together as shown below

(Finished) Your machine should now be ready for connecting power and testing. Please check that the Main AC voltage is correct (marked on the back of the machine)

*** NOTE! ***

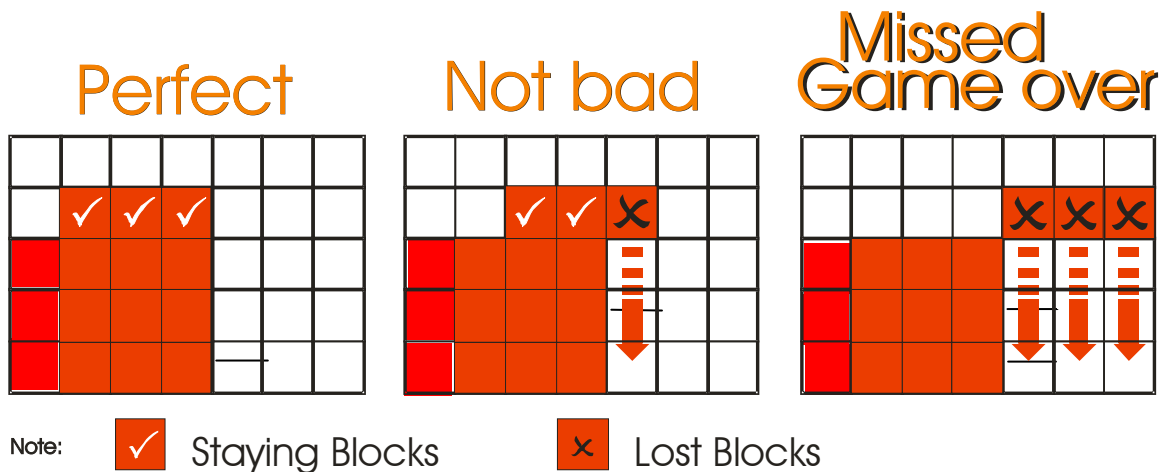
Make sure to have taken all cable ties off the crane gantry



HOW TO PLAY

PLAYERS TRY TO STACK THE BLOCKS VERTICALLY TO WIN PRIZES

- Pay to play.
- Press the Stack button to start a game.
- Press the Stack button to stop the moving blocks.
- Stop the moving blocks on top of the blocks in previous level.
- Game ends when the player fails to stop the moving blocks above any of the blocks in the previous level.
- The Player wins if the last block is stacked correctly on the last level the claw will then become active.
- Winning players use the Joy stick to maneuver the crane and drop the claw to “Grab” their prize. The claw remains active until a prize has been dropped into the vend area.



*** NOTE! ***

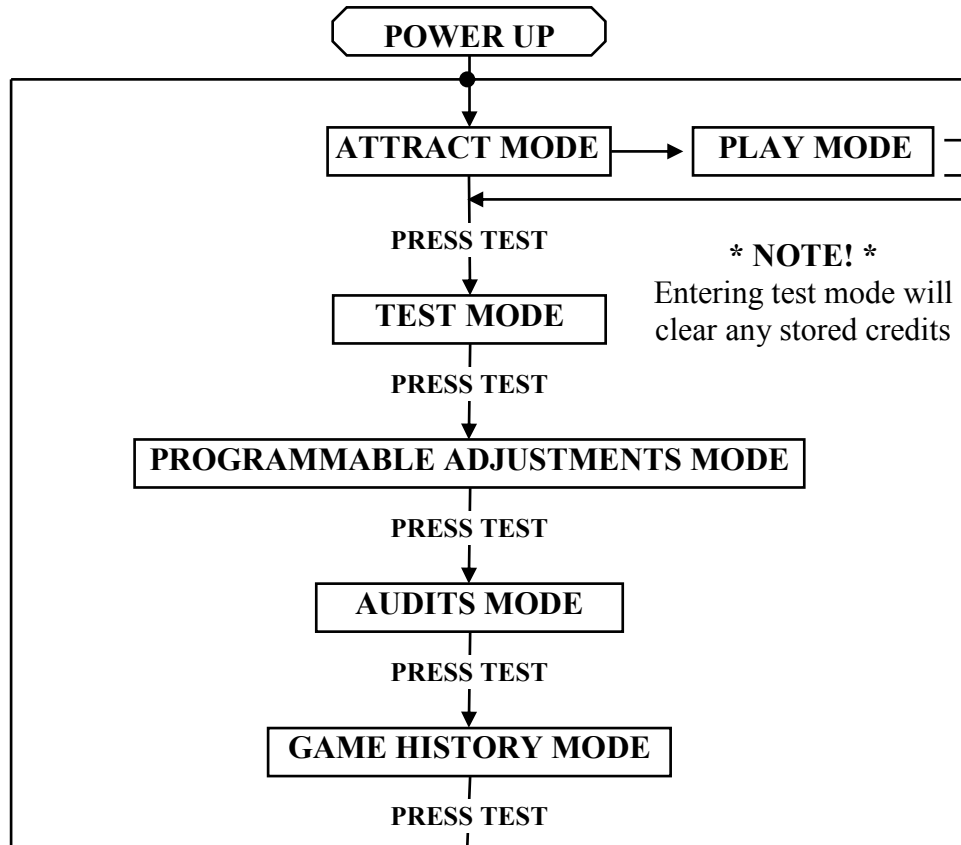
Toys Storage Height 300 mm (11”) x Width 1124 mm (43”) x Depth 712 mm (28”)



OPERATION

The “**Stack N Grab**” game has five operational modes: Attract mode, Play mode, Test mode, Programmable Adjustments Mode Audits Mode and History Mode.

OPERATIONAL DIAGRAM



ATTRACT MODE

The Attract mode provides a light and sound display, while the game is not being played. This feature is to attract potential customers to play the game. The attract mode sound can be turned on and off
(Refer to programmable adjustment page of this manual).

PLAY MODE

- “**Stack N Grab**” has two play modes. The Standard Coin Play mode, where a coin or coins are inserted to play the game, or Free Play where no coins are necessary.

COIN PLAY

- The Coin Play mode is entered from Attract mode, by inserting coins in any of the two coin slots on the front of the machine cabinet, then following the instructions in the “How to Play” section of this manual.

FREE PLAY

- The free play mode is entered from attract mode by holding the Service button for longer than five seconds, **FFEE** will be displayed on the 4-digit LED display. For a single free game, just press the Service button once. When issuing single free games in this manner, Prizes can be won as normal.



TEST MODE

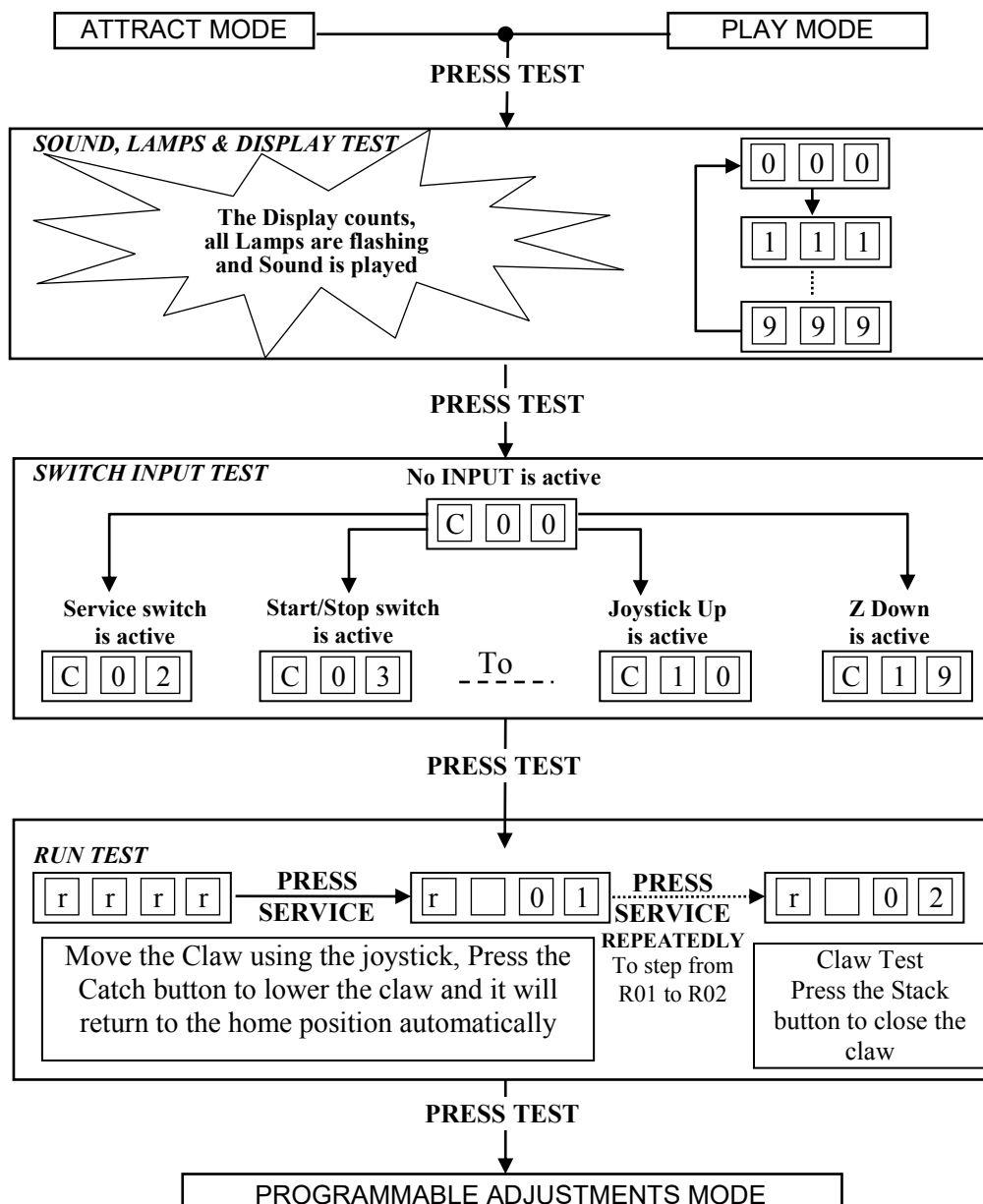
The “**Stack N Grab**” Test mode has Three Test Configurations allowing you to explore the functioning of the Sound, Light & Display and Game Switches and to allow an operational test of the Crane Claw (Refer to the Test Mode Diagram below).

The Test mode is also used for Clearing Game Errors. If there is an active error, its code will be displayed. To try to clear the error code, press the red test button once. The error can be by passed by quickly pressing the red test button twice.

* NOTE! *

- Entering Test Mode will CLEAR any CREDITS remaining in the game.
- If during test mode no ADJUSTMENTS or actions are made to the game for approximately four minutes, it will automatically RETURN to Attract Mode.

TEST MODE DIAGRAM





SOUND, LAMPS & DISPLAY TEST

- **ENTER** The Sound, Lamp & Display test is entered from Attract mode by pressing the test button once.

*** NOTE! ***

- If there is an active error displayed, press the red test button once to try and clear the error.
- If the error code will not clear, it can be bypass by quickly pressing the red test button twice.

DURING THE TEST:

- Game music and a voice over will be played.
 - The Door Indicator LEDs will light up in sequence.
 - The Credit display will count from 0000 to 9999 and then repeat.
 - The LED Playfield Display panel will run a test pattern sequence.
 - The Continue, Start/Stop and Select button lamps will flash on and off
- **EXIT** The Sound, Lamp & Display test is exited by pressing the test button. The next test will be switch test.



SWITCH TEST

- **ENTER** The Switch Test can be entered by pressing the Test button once while in the Sound, Light & display test or by pressing the Test button twice while in Attract mode, **C-XX** will be displayed on the 4-digit display where „XX“ is a number representing the switch that is active.

- **TESTING THE GAME SWITCHES**

All game switches have a code from C1 to C19 as tabled below. By activating any of the switches, their code will be displayed on the 4-digit display. In the normal condition with the crane in the home position, **C-15**, **C-16** and **C-19** will be active and shown in the 4- Digit display.

CODE	DISPLAY	SWITCH FUNCTION
C-0	C - 0 0	No Input Active
C-1	C - - 1	(Optional)Ticket/Capsule Switch Active
C-2	C - - 2	Service Switch Active
C-3	C - - 3	Stack Button Active
C-4	C - - 4	Coin 1 Switch Active
C-5	C - - 5	Coin 2 Switch Active
C-9	C - - 9	Prize Sensor Active
C-10	C - 1 0	Joystick Up Switch Active
C-11	C - 1 1	Joystick Down Switch Active
C-12	C - 1 2	Joystick Right Switch Active
C-13	C - 1 3	Joystick Left Active
C-14	C - 1 4	Joystick Button Switch Active
C-15	C - 1 5	Crane Mechanism Left Home Switch
C-16	C - 1 6	Crane Mechanism Forward Home Switch
C-17	C - 1 7	Crane Mechanism Back Home Switch
C-18	C - 1 8	Claw Down Position Switch
C-19	C - 1 9	Claw Up Position Home Switch

*** NOTE! ***

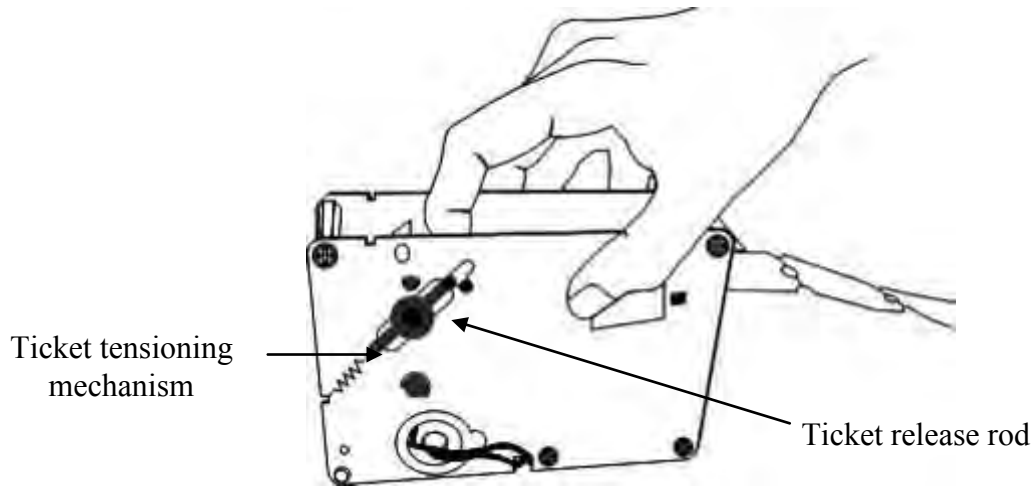
- Several switches can be simultaneously activated in Switch test. The display will then consecutively show their codes, indicating which switches are active. However, it is much easier to test the game switches individually.



■ TICKET DISPENSER NOTCH

(If optional Ticket or Capsule dispenser is fitted)

The Ticket Notch Switch (C1) can be activated or deactivated from the Ticket Feed Button on the Ticket Dispenser PCB or by manually pushing the tickets from the ticket holder through the dispenser after pulling the ticket release rod upwards



- **EXIT** The Switch Test is exited into Run Test Mode by pressing the Test Button once.

RUN TEST

- **ENTER** The Run Test can be entered by pressing the Test button once while in the Switch Test or by pressing the Test button three times while in Attract mode, **r r r r** will be displayed on the 4-digit display.

- **SELECT** The Service button is pressed once to start the run test mode. The credit display will indicate, **r - 0 1** for the first crane mechanism run test. The Service button is then pressed again to step to the second crane mechanism test r-02.

- **RUN** Use the joystick and the Stack button to operate the Crane Mechanism in the Run test.

- **EXIT** The Run Test is exited into Programmable Adjustments Mode by pressing the Test Button once.

■ RUN TEST 1 (r-01)

This test allows you to test the normal operation of the crane mechanism as it would work in a winning game, the joystick with move the mechanism in all directions and pushing the Stack button will drop the claw down and the claw will close and try to grab a prize and then return automatically to the home position this test can be repeated until you exit the test.

■ RUN TEST 2 (r-02)

The test is to test the claw only, by pushing the Stack button the claw with pull closed and stay closed while the button is pressed.

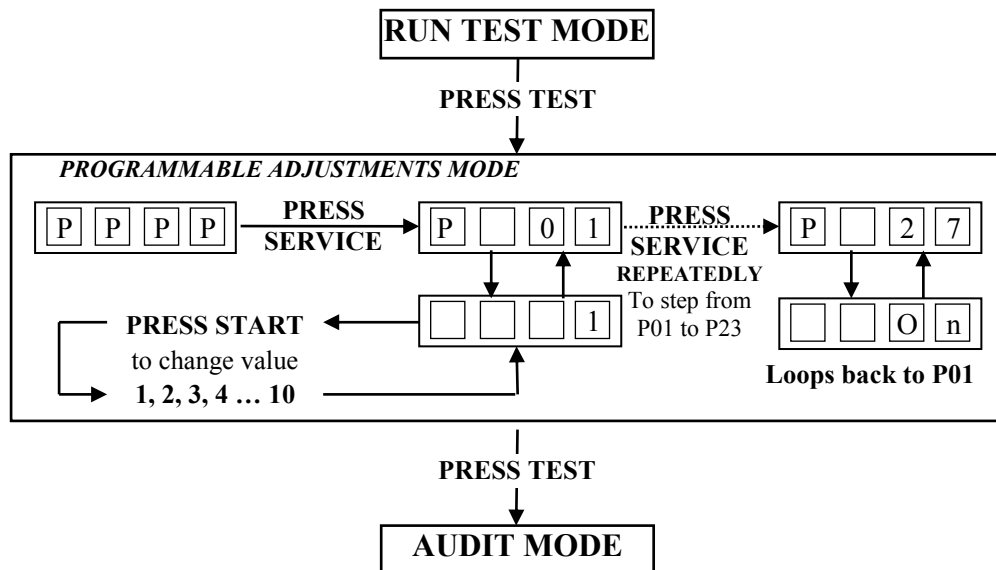


PROGRAMMABLE ADJUSTMENTS MODE

“Stack n Grab” has fifteen programmable adjustments that can be changed in this mode. They are P01 to P15 and their codes and values are displayed alternatively during the adjustment procedure.

Example: Code **P01** (*Number of Coins Mech 1*) is displayed as **P**□**0****1** and its value of **1** as □□□**1** on the 4-digit display.

PROGRAMMABLE ADJUSTMENTS MODE DIAGRAM



PROGRAMMABLE ADJUSTMENTS PROCEDURE

- **ENTER** The Programmable Adjustments Mode can be entered by pressing the Test button once while in the Run Test or by pressing the Test button four times while in Attract mode, **P P P P** will be displayed on the 4-digit credit display.
- **SELECT** The green Service button is pressed to step through each of the adjustment configurations, starting from the **P P P P** display, P01 being the first step, continuing through to P15, and then looping again from P01 to P15 until the mode is exited.
- **CHANGE** The Stack button is pressed to change the displayed value. The value can only be stepped up by using the Stack button, but the value will loop back to its minimum value the next step after its max value.

* NOTE! *

- Certain program adjustments have a fast adjustment feature. By holding the Start/Stop button down, the values step through quicker.
- **EXIT** The Programmable Adjustments mode is exited into Audits mode, by pressing the Test button once.



PROGRAMMABLE ADJUSTMENTS QUICK REFERENCE TABLE (V1.6.2)

CODE	PROGRAMMABLE ADJUSTMENTS	OPTIONAL VALUES	DEFAULT SETTINGS	FEATURES
P01	1 - 20	1,2,3,...20	1	Coin 1 – Coins / Credit
P02	1 - 10	1,2,3,...10	1	Coin 1 – Games / Credit
P03	ON or OFF	ON or OFF	OFF	Activate Multiple Bonus Pricing Coin slot 1
P3-1	OFF - 99	OFF,1,2,3,...99	OFF	Coin 1 Number Coins for Bonus Pricing level 1
P3-2	OFF - 99	OFF,1,2,3,...99	OFF	Coin1 Number of bonus credits on Pricing level 1
P3-3	OFF - 99	OFF,1,2,3,...99	OFF	Coin1 Number Coins for Bonus Pricing level 2
P3-4	OFF - 99	OFF,1,2,3,...99	OFF	Coin 1 Number of bonus credits on Pricing level 2
P3-5	OFF - 99	OFF,1,2,3,...99	OFF	Coin 1 Number Coins for Bonus Pricing level 3
P3-6	OFF - 99	OFF,1,2,3,...99	OFF	Coin 1 Number of bonus credits on Pricing level 3
P04	1 - 20	1, 2, 3...20	1	Coin 2 – Coins / Credit
P05	1 - 10	1, 2, 3...10	1	Coin 2 – Games / Credit
P06	OFF or ON	ON or OFF	OFF	Activate Multiple Bonus Pricing Coin slot 2
P6-1	OFF - 99	OFF,1,2,3,...99	OFF	Coin 2 Number Coins for Bonus Pricing level 1
P6-2	OFF - 99	OFF,1,2,3,...99	OFF	Coin 2 Number of bonus credits on Pricing level 1
P6-3	OFF - 99	OFF,1,2,3,...99	OFF	Coin 2 Number Coins for Bonus Pricing level 2
P6-4	OFF - 99	OFF,1,2,3,...99	OFF	Coin 2 Number of bonus credits on Pricing level 2
P6-5	OFF - 99	OFF,1,2,3,...99	OFF	Coin 2 Number Coins for Bonus Pricing level 3
P6-6	OFF - 99	OFF,1,2,3,...99	OFF	Coin 2 Number of bonus credits on Pricing level 3
P07	ON or OFF	ON or OFF	ON	Attract Mode sound
P08	3- 5	3- 5	3	(Read Only)Game Difficulty Cube Speed
P9	1 - 200	1, 2,30, 35,200	10	Average Games Per Win (AGW)
P10	ON or OFF	ON or OFF	OFF	Prizes dispensed when in free play
P11	ON or OFF	ON or OFF	ON	Attract Mode Display Animation strobes
P12	1...4	1,2,3,4	1	Error Message options
P13	0 – 15	0,3,5,...15	3	Crane Prize selection Time Out (Minutes)
P14	0 – 20	1,2,3,...20	0	Number of Mercy Tickets/Capsules (optional)
P15	0 – 2	0, 1 or 2	0	Mercy Tickets/Capsule Option



PROGRAMMABLE ADJUSTMENTS DETAILED

■ P01 = COIN 1: NUMBER OF COINS PER CREDIT

(Default 01) (Adjustable 1 – 20)

This sets the number of coins that need to be inserted into coin mechanism 1, for each credit. It can be set between 1 to 20 coins for one credit. The default setting is “1”.

■ P02 = COIN 1: NUMBER OF GAME PLAYS PER CREDIT

(Default 01) (Adjustable 1 – 10)

This sets the number of games for each credit inserted into coin mechanism 1. It can be set between 1 to 10 plays for each credit. The *default* setting is “1”.

■ P03 = COIN 1: ACTIVATE MULTIPLE BONUS PRICING

(Default OFF) (Adjustable ON – OFF)

Note: Settings P 03 and P 03-1 thru to P03-6 are only used for the setting of Bonus credit levels e.g. \$0.50c/1 play, \$1/3plays, \$2/7plays, \$5/20 plays

This turns on the multiple bonus credit system and activates the settings for up to 3 bonus levels on coin mechanism 1. It can be set to ON or OFF. The *default* setting is “OFF” this means the multiple bonuses is disabled, if the setting is changed to ON the multiple bonus setting will be active and open the next sub-menu **P03-1** and so on.

■ P03 - 1 = COIN 1: NUMBER OF COINS REQUIRED TO REACH BONUS CREDIT LEVEL 1

(Default OFF) (Adjustable OFF – 99)

This sets the number of coins (or Bill Acceptor pulses) that need to be inserted into coin mechanism 1 to reach the bonus credit level 1. It can be set to either OFF for no bonus or 1 to 99 coins, (OFF=No bonus), the *default* setting is “OFF” this means that the **P03-2** will not open

Examples	(Base price \$0.25c)	(Base Price \$0.50c)	(Base Price \$0.50c)	(Base Price \$1.00)
P Setting Adjustment	1 play \$ 0.25c 3 plays \$ 0.50c 7 plays \$ 1.00 (\$0.25c coins or DBA set on \$0.25c pulses)	1 play \$ 0.50c 3 plays \$ 1.00 7 plays \$ 2.00 (\$0.25c coins or DBA set on \$0.25c pulses)	1 play \$ 0.50c 3 plays \$ 1.00 8 plays \$ 2.00 22 plays \$ 5.00 (\$0.25c coins or DBA set on \$0.25c pulses)	1 play \$ 1.00 3 plays \$ 2.00 8 plays \$ 5.00 18 plays \$ 10.00 (\$0.25c coins or DBA set on \$0.25c pulses)
P01 / P04	1	2	2	4
P02 / P05	1	1	1	1
P03 / P06	ON	ON	ON	ON
P3-1 / P6-1	2	4	4	8
P3-2 / P6-2	1	1	1	1
P3-3 / P6-3	4	8	8	20
P3-4 / P6-4	3	3	4	3
P3-5 / P6-5	OFF	OFF	20	40
P3-6 / P6-6	OFF	OFF	12	8



■ **P03 - 2 = COIN 1: NUMBER OF BONUS CREDITS GIVEN AT BONUS LEVEL 1**

(Default OFF) (Adjustable OFF – 99)

This sets the number of bonus credits that are given when credit Level 1 is reached. This Bonus amount is the **additional** number of credits required above the **base price**. It can be set to either OFF, or between 1 to 99 bonus credits; the *default* setting is “OFF” this mean that the **P03-3** will not open.

Note: The **Base Price** is the normal price setting for one game.

e.g. If the game is set for \$0.25c/1play then the base price is \$0.25c, if the game is set for \$0.50c/1play then the base price is \$0.50c, if the game is set for \$1.00/1play then the base price is \$1.00.

■ **P03 - 3= COIN 1: NUMBER OF COINS REQUIRED TO REACH BONUS CREDIT LEVEL 2**

(Default OFF) (Adjustable OFF – 99)

This sets the number of coins (or Bill Acceptor pulses) that is needed to be inserted into coin mechanism 1 to reach the bonus credit level 2. It can be set to OFF for no bonus or between 1 to 99 coins, but the setting value must be higher than setting value of **P03-1**, the *default* setting is “OFF” and if set to OFF this means that the **P03-4** will not open.

■ **P03 - 4 = COIN 1: NUMBER OF BONUS CREDITS GIVEN AT BONUS LEVEL 2**

(Default OFF) (Adjustable OFF – 99)

This sets the number of bonus credits that are given when credit Level 2 is reached. This Bonus amount is the **additional** number of credits required above the **base price**. It can be set to either OFF, or between 1 to 99 bonus credits; the *default* setting is “OFF” this mean that the **P03-5** will not open.

■ **P03 - 5= COIN 1: NUMBER OF COINS REQUIRED TO REACH BONUS CREDIT LEVEL 3**

(Default OFF) (Adjustable OFF – 99)

This sets the number of coins (or Bill Acceptor pulses) that is needed to be inserted into coin mechanism 1 to reach the bonus credit level 3. It can be set to OFF for no bonus or between 1 to 99 coins, but the setting value must be higher than setting value of **P03-3**, the *default* setting is “OFF” and if set to OFF this mean that the **P03-6** will not open.

■ **P03 - 6 = COIN 1: NUMBER OF BONUS CREDITS GIVEN AT BONUS LEVEL 3**

(Default OFF) (Adjustable OFF – 99)

This sets the number of bonus credits that are given when credit Level 3 is reached. This Bonus amount is the **additional** number of credits required above the **base price**. It can be set to either OFF, or between 1 to 99 bonus credits; the *default* setting is “OFF”



■ **P04 = COIN 2: NUMBER OF COINS PER CREDIT**

(Default 01) (Adjustable 1 – 20)

This sets the number of coins that need to be inserted into coin mechanism 2, for each credit. It can be set between 1 to 20 coins for one credit. The default setting is “1”.

■ **P05 = COIN 2: NUMBER OF GAME PLAYS PER CREDIT**

(Default 01) (Adjustable 1 – 10)

This sets the number of games for each credit inserted into coin mechanism 2. It can be set between 1 to 10 plays for each credit. The *default* setting is “1”.

■ **P06 = COIN 2: ACTIVATE MULTIPLE BONUS PRICING**

(Default OFF) (Adjustable ON – OFF)

Note: Settings P 06 and P 06-1 thru to P06-6 are only used for the setting of Bonus credit levels e.g. \$0.50c/1 play, \$1/3plays, \$2/7plays, \$5/20 plays

This turns on the multiple bonus credit system and activates the settings for up to 3 bonus levels on coin mechanism 2. It can be set to ON or OFF. The *default* setting is “OFF” this mean the multiple bonuses is disabled, if the setting change to ON the multiple bonus setting will be active and open the next sub-menu **P06-1** and so on.

■ **P06 - 1 = COIN 2: NUMBER OF COINS REQUIRED TO REACH BONUS CREDIT LEVEL 1**

(Default OFF) (Adjustable OFF – 99)

This sets the number of coins (or Bill Acceptor pulses) that need to be inserted into coin mechanism 2 to reach the bonus credit level 1. It can be set to either OFF for no bonus or 1 to 99 coins, (OFF=No bonus), the *default* setting is “OFF” this means that the **P06-2** will not open

■ **P06 -2 = COIN 2: NUMBER OF BONUS CREDITS GIVEN AT BONUS LEVEL 1**

(Default OFF) (Adjustable OFF – 99)

This sets the number of bonus credits that are given when credit Level 1 is reached. This Bonus amount is the **additional** number of credits required above the **base price**. It can be set to either OFF, or 1 to 99 bonus credits; the *default* setting is “OFF” this mean that the **P06-3** will not open.

Note: The **Base Price** is the normal price setting for one game.

e.g. If the game is set for \$0.25c/1play then the base price is \$0.25c, if the game is set for \$0.50c/1play then the base price is \$0.50c, if the game is set for \$1.00/1play then the base price is \$1.00,

■ **P06 – 3= COIN 2: NUMBER OF COINS REQUIRED TO REACH BONUS CREDIT LEVEL 2**

(Default OFF) (Adjustable OFF – 99)

This sets the number of coins (or Bill Acceptor pulses) that is needed to be inserted into coin mechanism 2 to reach the bonus credit level 2. It can be set to OFF for no bonus or between 1 to 99 coins, but the setting value must be higher than setting value of **P06-1**, the *default* setting is “OFF” and if set to OFF this means that the **P06-4** will not open.



■ **P06 - 4 = COIN 2: NUMBER OF BONUS CREDITS GIVEN AT BONUS LEVEL 2**

(Default OFF) (Adjustable OFF – 99)

This sets the number of bonus credits that are given when credit Level 2 is reached. This Bonus amount is the **additional** number of credits required above the **base price**. It can be set to either OFF, or 1 to 99 bonus credits; the *default* setting is “OFF” this mean that the **P06-5** will not open.

■ **P06 – 5 = COIN 2: NUMBER OF COINS REQUIRED TO REACH BONUS CREDIT LEVEL 3**

(Default OFF) (Adjustable OFF – 99)

This sets the number of coins (or Bill Acceptor pulses) that is needed to be inserted into coin mechanism 2 to reach the bonus credit level 3. It can be set to OFF for no bonus or between 1 to 99 coins, but the setting value must be higher than setting value of **P06-3**, the *default* setting is “OFF” and if set to OFF this mean that the **P06-6** will not open.

■ **P06 -6 = COIN 2: NUMBER OF BONUS CREDITS GIVEN AT BONUS LEVEL 3**

(Default OFF) (Adjustable OFF – 99)

This sets the number of bonus credits that are given when credit Level 3 is reached. This Bonus amount is the **additional** number of credits required above the **base price**. It can be set to either OFF, or 1 to 99 bonus credits; the *default* setting is “OFF”.

■ **P07 = ATTRACT MODE SOUND**

(Default ON) (Adjustable ON or OFF)

This adjustment turns the *attract mode sound* **ON** or **OFF**. This is the sound and music that the game generates to attract customers when it is not being played. The music will cycle approximately every 3 minutes.

■ **P08 = GAME DIFFICULTY (Cube Speed)**

(Default 3) (Non Adjustable 3 - 5)

This option displays the current setting of the *Game Difficulty which varies the speed that the blocks move from side to side. This setting is “read only” as the setting is adjusted automatically by the game software.*

■ **P09 = AVERAGE GAMES PER WIN (AGW)**

(Default 10) (Adjustable 1 – 200)

This option is for adjusting the game payout and sets the *Average Games played per Prize Won*. The game sets the game difficulty to achieve a player win ratio that will average very close to the (AGW). Example an AGW setting 20 means on average a prize will be won for every 20 games played. The setting is adjustable from 1 to 200. With adjustment from 1-30 in steps of 1 and from 30-200 in steps of 5. The *default* setting is 10, or one win every 10 Games which suitable for \$4 Plush on \$0.50 per play



■ **P10 = PRIZES IN FREE PLAY MODE**

(Default OFF) (Adjustable ON or OFF)

This setting controls whether or not the *game dispenses prizes* in free play mode. The options are **ON** or **OFF**.

■ **P11 = ATTRACT ANIMATION (STROBING) DISPLAY**

(Default ON) (Adjustable ON or OFF)

This setting controls whether or not the games attract mode animation will strobe. When set to ON, the game will run the attract animation which includes a small amount of strobe effect. If set to OFF, the game will skip this part of the attract animation.

■ **P12 = ERROR MESSAGE OPTION**

(Default 1) (Adjustable 1 - 4)

This adjustment sets the way error messages are handled. The game can play a voice over for an error, or display the error on the small 4 digit display or both.

Setting	Voice Over	4 Digit Display
1	Played	Displayed
2	Played	Errors will only display when the test button is pressed and will try clear when the test button is pressed again
3	Not Played	Displayed
4	Not Played	Errors will only display when the test button is pressed and will try clear when the test button is pressed again

■ **P13 = PRIZE SELECTION TIME OUT**

(Default 3) (Adjustable 0, 3 – 15)

This adjustment sets the crane time out (in minutes) for a player to pick up a Toy/Prize after winning. Example, when set to 3 means a winning player has 3 minutes to select and pick up a Toy/Prize. When the timeout has been reached, the game will go to Err4 (prize deployment Error) A setting of 0 means there is no time out.

Note: If the player fails to pick up a Toy/Prize three times in a row the crane will automatically run a self test by lowering the claw into the prize chute to make sure the prize sensor is working OK.

NOTE! To Clear the Err 4 Press Red Test button once.



■ **P14 = NUMBER OF MERCY TICKETS / CAPSULES**

ADJUSTMENT (default 0) (Adjustable 0 – 20)

This option adjusts the number of mercy tickets or capsules paid out if the optional ticket or capsule dispenser is fitted. If there is no optional Ticket or Capsule Dispenser fitted the setting should be 0

■ **P15 == MERCY SYSTEM MODE ADJUSTMENT**

(Default 1) (Adjustable 1 – 2)

This option adjusts the way that mercy tickets or capsules are paid out if the optional ticket or capsule dispenser is fitted. See **P14** for setting the number of mercy tickets or capsules that will be dispensed.

1. Mercy tickets / capsules are paid if no prize is won. Optional ticket / capsule dispenser must be fitted.
2. Mercy tickets / capsules are paid on every game credit, regardless if prizes are won or not. Optional ticket / capsule dispenser must be fitted



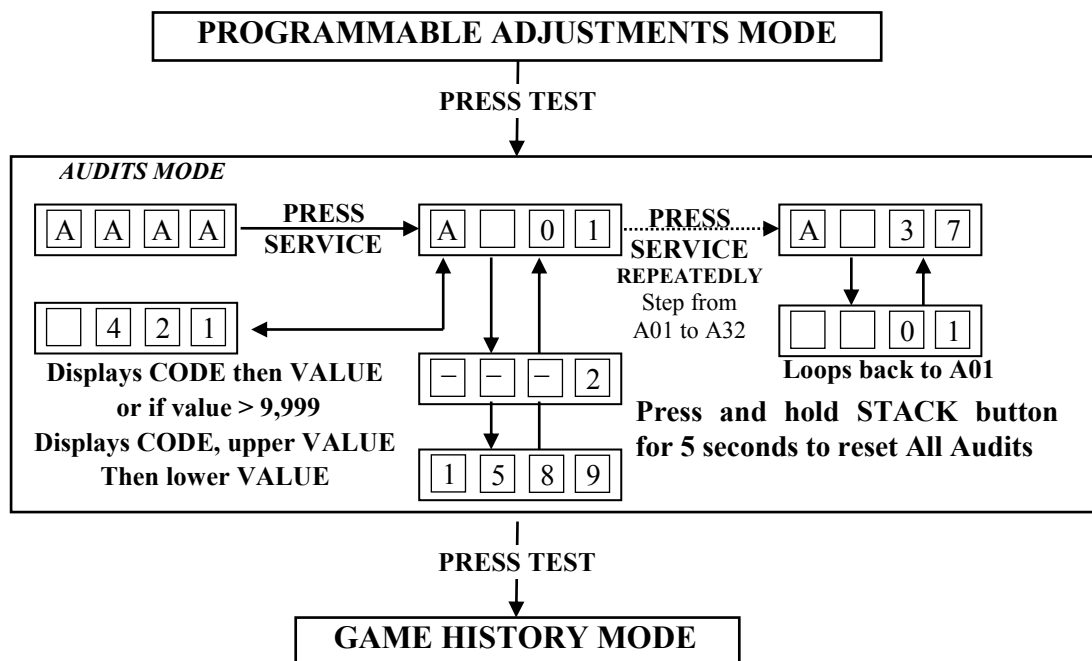
AUDITS MODE

The Audits Mode allows the operator to view statistics in all areas of the Game Play. This enables the operator to make calculated adjustments and “Fine Tune” the machine to maximize earning potential. The Audits mode stores bookkeeping of the games processed since the last game audits reset. While in this mode, the game audits can also be reset to zero.

The Stack N Grab has thirty seven Audits that can be viewed in this mode. They are A01 to A37 and their codes and values are displayed alternatively during the Audit Mode.

Example: Code **A01** will be displayed as **A** **0** **1** and a value of **421** as **4** **2** **1** on the 4-digit display.
Or it will display large values like **21589** as **- - - 2** and **1 5 8 9** on the 4-digit display.

AUDITS MODE DIAGRAM



* NOTE! *

- For Audit values that are greater than 4 digits the audits' values will be displayed in two steps.
- The first number, which is displayed as **- - - 2**, has leading dash symbols
- The second number, which is displayed as **1 5 8 9**, which has no dash symbols
- In this example the final value is 21,589



AUDIT PROCEDURE

- **ENTER** The Audits mode is entered from Programmable Adjustments mode by pressing the Test button once or from Attract mode by pressing the Test button five times. **A A A A** Will be displayed on the 4-digit display.
- **SELECT** The green Service button is pressed for advancing each step through the set of audits configurations, starting from the **A A A A** display, A01 being the first step, continuing through to A37, and then looping again from A01 to A37 until the mode is exited.
- **RESET** The entire set of user audits can be reset during any of the audit configurations, by holding the Stack button for longer than 5 seconds. The displays will be cleared while still holding the button pressed and will return to the same audit step after releasing the button. The value of all audits will be reset to “00 000”.
- **EXIT** The Audits mode is exited into Game History mode, by pressing the Test button once.

* NOTE! *

- **ALL** Audits will **STOP INCREMENTING** when the “Total Number of Games Played”, audit A-07, reaches 60,000.
- To restart the audits they must be reset to 0000 by holding The Start button for longer than 5 seconds while in Audits mode.



AUDITS QUICK REFERENCE TABLE

CODE	DISPLAY	AUDIT FUNCTION
A01	A - 0 1	Total Coins in Mechanism 1
A02	A - 0 2	Total Coins in Mechanism 2
A03	A - 0 3	Total Number of Service Credit
A04	A - 0 4	Average Number of Games Played Per Prize Win
A05	A - 0 5	Average Level Reached Per Game
A06	A - 0 6	Total Number of Major Prizes Won
A07	A - 0 7	Total Number of games played
A08	A - 0 8	Total Number Games reach level 1
A09	A - 0 9	Total Number Games reach level 2
A10	A - 1 0	Total Number Games reach level 3
A11	A - 1 1	Total Number Games reach level 4
A12	A - 1 2	Total Number Games reach level 5
A13	A - 1 3	Total Number Games reach level 6
A14	A - 1 4	Total Number Games reach level 7
A15	A - 1 5	Total Number Games reach level 8
A16	A - 1 6	Total Number Games reach level 9
A17	A - 1 7	Total Number Games reach level 10
A18	A - 1 8	Total Number of Mercy Games
A19	A - 1 9	Total Prize Counter (non-resettable)
A20	A - 2 0	Total Game Plays (non-resettable)
A21	A - 2 1	Coin 1 Counter (non-resettable)
A22	A - 2 2	Coin 2 Counter (non-resettable)
A23	A - 2 3	(Manufacture Audit)
A24	A - 2 4	(Manufacture Audit)
A25	A - 2 5	(Manufacture Audit)
A26	A - 2 6	(Manufacture Audit)
A27	A - 2 7	(Manufacture Audit)
A28	A - 2 8	(Manufacture Audit)
A29	A - 2 9	(Manufacture Audit)
A30	A - 3 0	(Manufacture Audit)
A31	A - 3 1	(Manufacture Audit)
A32	A - 3 2	(Manufacture Audit)
A33	A - 3 3	(Manufacture Audit)
A34	A - 3 4	(Manufacture Audit)
A35	A - 3 5	(Manufacture Audit)
A36	A - 3 6	(Manufacture Audit)
A37	A - 3 7	(Manufacture Audit)



AUDITS DETAILED

■ A01 = TOTAL COINS IN MECHANISM 1

This Audit displays the *total number of coins* inserted into coin mechanism 1 since the audits were last cleared.

■ A02 = TOTAL COINS IN MECHANISM 2

This Audit displays the *total number of coins* inserted into coin mechanism 2 since the audits were last cleared.

■ A03 = TOTAL NUMBER OF SERVICE CREDITS

This Audit displays the total number of *Service Credits* since the audits were last cleared. This records the number of credits given by pressing the Service Button on the service panel.

■ A04 = AVERAGE GAMES PER WIN (AGW)

This Audit displays the current *Average number of games played for every prize won* since the audits were last cleared.

■ A05 = AVERAGE LEVEL REACHED IN THE GAME

This Audit is monitors the players skill level and the game difficulty, it displays the *Average Level that players reach* since the audits were last cleared.

■ A06 = TOTAL NUMBER OF PRIZES PAID OUT

This Audit displays the *total number of Prizes Won*, since the audits were last cleared.

■ A07 = TOTAL GAMES PLAYED

This Audit displays the *total number of Games Played* since the audits were last cleared.

* NOTE! *

- ALL Audits will **STOP INCREMENTING** when the “Total Number of Games Played”, audit A-07, reaches 60,000.
- To restart the audits they must be reset to 00 000 by holding The Start button for longer than 5 seconds while in audits mode.



■ **A08 to A17**

TOTAL NUMBER OF GAMES ENDING ON LEVELS 1 to 10

These Audits display the *total number of games ending on level* number 1 through to 10 on this machine since the audits were last cleared.

■ **A18 = TOTAL NUMBER OF MERCY GAMES**

This Audit displays the *total number of “Mercy Games”* that have been played since the audits were last cleared. Note: this audit is only active if the optional Capsule or Ticket dispenser is fitted and the P-14 Mercy Option is active

■ **A19 = TOTAL NUMBER OF PRIZES PAID OUT
(Non Resettable)**

This is a non resettable counter that displays the total number of Prizes paid out.
NOTE: this audit cannot be reset.

■ **A20 = TOTAL GAMES PLAYED (Non Resettable)**

This is a non resettable counter that displays the total number of Games Played.
NOTE: this audit cannot be reset.

■ **A21 = COIN 1 COUNTER (Non Resettable)**

This is a non resettable counter that displays the total number of coins inserted into coin mechanism 1. **NOTE: this audit cannot be reset.**

■ **A22 = COIN 2 COUNTER (Non Resettable)**

This is a non resettable counter that displays the total number of coins inserted into coin mechanism 2. **NOTE: this audit cannot be reset.**

■ **A23 to A37 = MANUFACTURERS AUDITS ONLY**

These are Manufacturer Audits only and serve no useful function for the operator of this game.

*** NOTE! ***

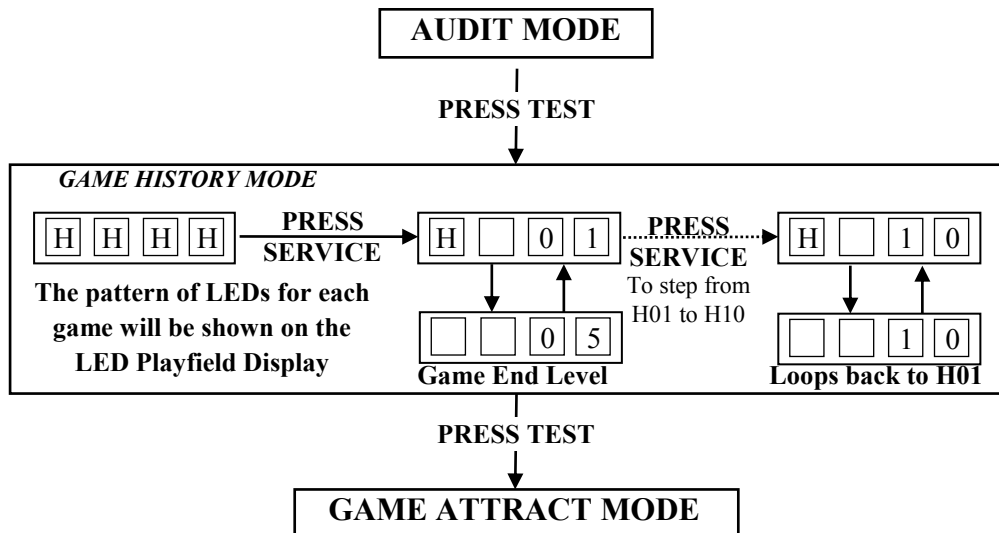
- **LAI Games** Customer Support may request from the operator the values of these Manufacturers audits, to help with any service issues.



GAME HISTORY MODE

By using the Game History Mode the operator can view the results of the last 10 games played. This enables the operator to verify player's game results and verify the win / lose pattern on the LED Playfield Display.

GAME HISTORY MODE DIAGRAM



*** NOTE! ***

- Score Histories will be erased if the game is switched off then on. Empty score histories show as [] [] [] [] on the 4-digit display

GAME HISTORY PROCEDURE

- ENTER** The Game History mode is entered from Audits mode by pressing the Test button once or from Attract mode by pressing the Test button six times. **H H H H** Will be displayed on the 4-digit display.
- SELECT** The green Service button is pressed for advancing each step through the set of Game Histories, starting from the **H H H H** display, H01 being the first step, continuing through to H10, and then looping again from H01 to H10 until the mode is exited.
- EXIT** The Game History mode is exited into Game Attract mode, by pressing the Test button once.



GAME HISTORY QUICK REFERENCE TABLE

CODE	DISPLAY	HISTORY RESULTS
H01	H - 0 1	Level Ending & LED Pattern for Very Last Game Played
H02	H - 0 2	Level Ending & LED Pattern for 2 nd Last Game Played
H03	H - 0 3	Level Ending & LED Pattern for 3 rd Last Game Played
H04	H - 0 4	Level Ending & LED Pattern for 4 th Last Game Played
H05	H - 0 5	Level Ending & LED Pattern for 5 th Last Game Played
H06	H - 0 6	Level Ending & LED Pattern for 6 th Last Game Played
H07	H - 0 7	Level Ending & LED Pattern for 7 th Last Game Played
H08	H - 0 8	Level Ending & LED Pattern for 8 th Last Game Played
H09	H - 0 9	Level Ending & LED Pattern for 9 th Last Game Played
H10	H - 1 0	Level Ending & LED Pattern for 10 th Last Game Played



ERRORS AND TROUBLESHOOTING

If the Game microprocessor detects any problems with the operation of the game, an Error will be displayed on the 4-digit display and the machine will play a voice message. “Please Call the Attendant/ be a Winner”. Some error Messages will only be displayed when test mode is entered. Errors are displayed on the displays as **E r r X**, where „X” is the error number. There are five error messages for *Stack N Grab*, listed as follows:

ERROR CODE QUICK REFERENCE TABLE

CODE	ERROR DESCRIPTION	SOLUTION
Err1	TICKET/CAPSULE No tickets/capsule, Jammed	Check the ticket/capsule Check the sensor/switch Check the Drive output to ticket/capsule
Err2	STACK BUTTON JAMMED , active for longer than 30 seconds	Check Button function using switch test Check the NO/NC connection
Err3	EEPROM ERROR Problem with on-board EEPROM	The main MCU is getting errors reading the EEPROM (24C16 IC on MCU).
Err4	PRIZE DEPLOYMENT ERROR Claw, Prize Sensor, Crane PCB or No Prizes. This is a hard Error <i>Test Mode can be access by pressing Test button.</i>	Check the Claw or Crane switches Check the prize sensor Check the Crane Control PCB Fill the cabinet with toys Check the Claw and Motor Fuses
Err5	PRIZE SENSOR BLOCKED or PRIZE SENSOR FAULTY	Clear the blockage from in front of the prize sensors Test Prize Sensors and replace if faulty
Err8	THE CRANE HAS NOT RETURNED TO THE HOME POSTION or HOME POSITION SWITCH NOT WORKING	Check the crane mechanism home position switch, make sure the connection is OK Check the motor operation in the Crane Gantry



TROUBLESHOOTING GAME ERRORS

■ CLEARING GAME ERRORS

Game errors can be cleared, by pushing the test button ONCE. The game will try and check if the error is fixed. If the reason for the error is fixed, the game will continue as normal. If the error is not fixed, the error will remain on the display. For a Hard Error (Err4) Power OFF and ON the game will clear the error.

■ Err1 – TICKET/CAPSULE ERROR

This error is usually displayed if the optional ticket /capsule dispenser does not function properly, or if tickets/capsules are jammed. Check the ticket/capsule dispenser is full, Check the ticket/capsule sensor/switch and make sure they are working properly, you can check this with your hand on the Capsule dispenser to make sure the sensor/switches are working also make sure the micro switch wiring is connected to the Normal Open and the Common contact of the micro switch. For a Ticket dispenser a ticket can be pushed in and out of the sensor to test it Use the switch test to help check the sensor/switch, an active switch will display as C1 in switch test.. Use a Digital Multimeter to check the voltage drive from the main CPU output to the motor or ticket/capsule connector.

■ Err2 – STACK BUTTON JAMMED

This error is usually displayed if the STACK button is active or jammed on for longer than 30 seconds. Check the mechanical operation of the Stack button and also the micro switch. Lastly make sure the micro switch wiring is connected to the Normal Open and the Common contact of the micro switch. Use the Switch Test to help check the Stack button, an active/pushed button will be display as **C3**,

■ Err3 – EEPROM ERROR

This Error means that the CPU cannot read the EEPROM, or is receiving errors during communication with the EEPROM (The 23C16 IC on the main MCU PCB). This could cause problems with the game audits and program settings. The first thing to do is try to switch the machine ON and OFF at least 2 times. If the message still appears then replace the EEPROM IC (Atmel 24C16) on the CPU PCB with the new EEPROM. If there is still an error message, this could be a problem with the game audits and program. If this error cannot be cleared, please send your main MCU PCB to the nearest authorized **LAI Games** Distributor for repair.

■ Err4 – PRIZE DEPLOYMENT ERROR

This error is displayed when the Claw is faulty, the Prize Sensor is not functioning either during the start up test or in game play and/or there are no Toys inside the cabinet and the game times out. This is a hard Error and will stop the game operating but you can still access the test mode by pressing the test button twice for fault finding the problem.

Note: Err4 can also occur when the claw rope winds around the pulley incorrectly and the claw is held up in the gantry. To clear this simply press and hold the Up switch inside the gantry for the claw to be released then redo the rope and wind it correctly. (Refer to the “Rope Winding For the Claw” diagram in the Service Section of this manual.)



■ **Err5 – PRIZE SENSOR BLOCKED or PRIZE SENSOR FAULTY**

This error occurs if the prize sensor is blocked or a prize is jammed in the prize chute, blocking the infrared beam of the prize sensor for longer than 5 seconds. Clear the blockage in front of the sensor; re adjust the sensor by turning the sensitivity screw in the sensor body if required.

■ **Err8 – HOME SWITCH GANTRY ERROR**

This error occurs if the Gantry home position switch is not functioning or the crane gantry has not properly returned to the home position, (Front/left position). Check the switch condition and operation, the motor and the switch stopper and check the connection and the wiring to this switch.



FUSE INFORMATION

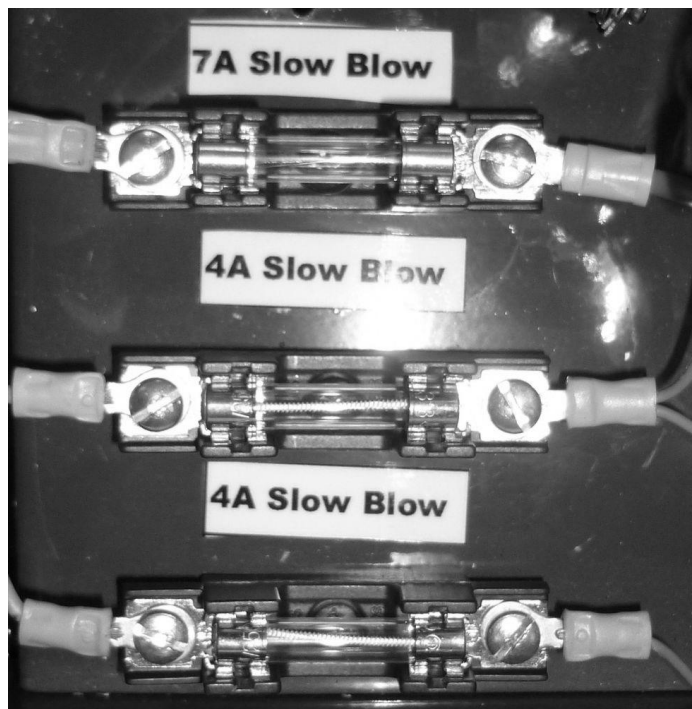
*** WARNING! ***

Always turn **OFF** Mains power and unplug the game, before replacing any fuses.

- **MAIN AC SUPPLY FUSE (1 x 6 AMP FAST BLOW, M205 TYPE)**
This fuse is for the main AC supply and is situated in the IEC mains input socket.
- **MCU POWER FUSE (1 x 1.5 AMP FAST BLOW, M205 TYPE)**
This fuse is for the power supply to the MCU PCB.
- **MCU CONTROL FUSES (2 x 5 AMP FAST BLOW, M205 TYPE)**
These fuses are for the DC transistor drivers on the MCU PCB
- **3 LED PLAYFIELD DISPLAY CONTROLLER FUSES
(3 x 2.5 AMP FAST BLOW, M205 TYPE)**
This fuse is for the +5VDC on the three LED Playfield Display PCBs
- **CLAW and MOTOR FUSES (2 x 4 AMP SLOW BLOW AND 7 AMP
SLOW BLOW, 3AG TYPE)**
- These fuses are for the Crane Mechanism Motors 48VDC (2 x 4A Slow Blow)
and for the Claw Coil 48 VDC 7A Slow Blow

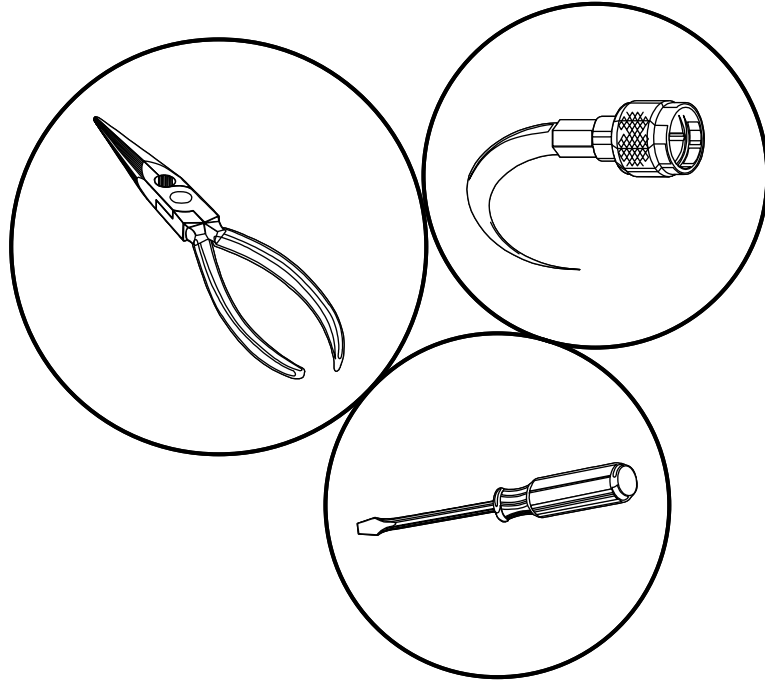
*** CAUTION! ***

Do Not Use any fuse that does not meet the specified rating.





SECTION A: SERVICE INSTRUCTIONS



BE SURE TO READ THE FOLLOWING
Carefully before servicing this machine

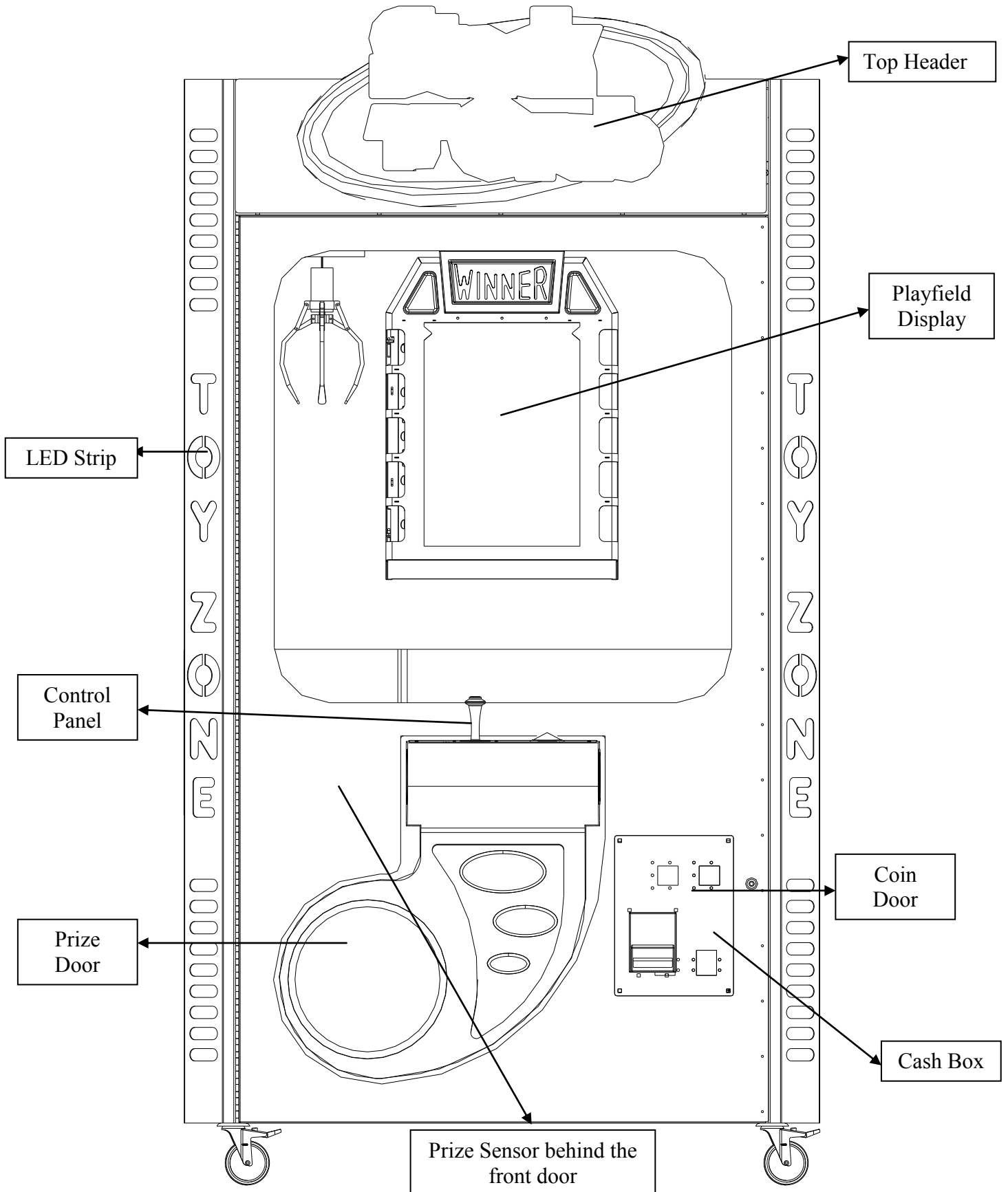




LOCATING AND ACCESSING PARTS

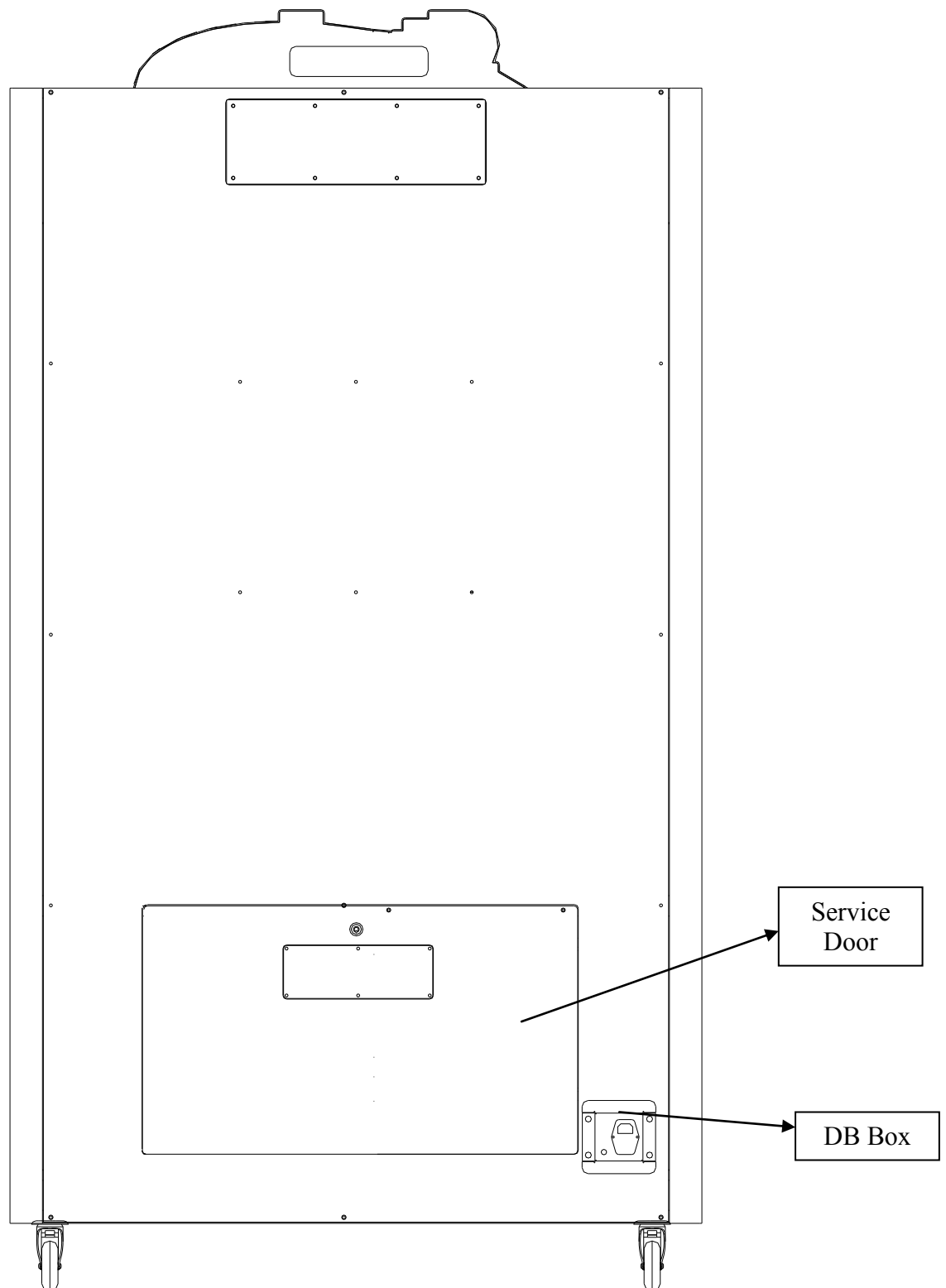
PARTS LOCATION DIAGRAM

As viewed from front



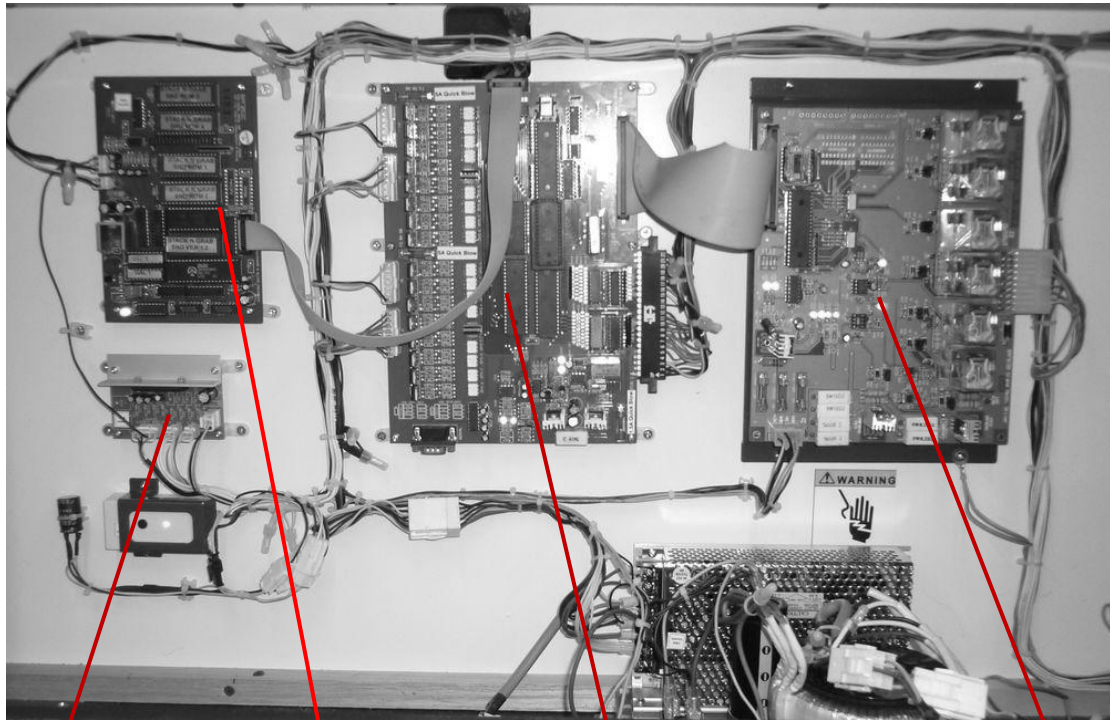


As viewed from back





As viewed from back continued,



BAFB29 Amplifier PCB

BAFB66A Main CPU PCB

BAFB158 Claw Control PCB

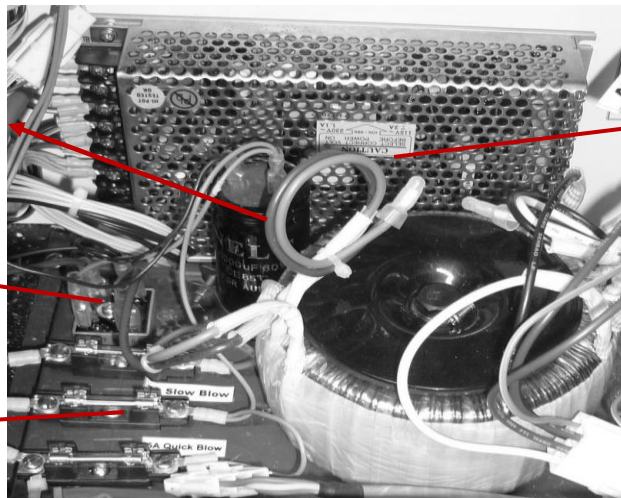
BAFB52C Sound ROM PCB

10.000 μ f/80 V
Capacitor

32 Amp Bridge
Diode

Fuse Location

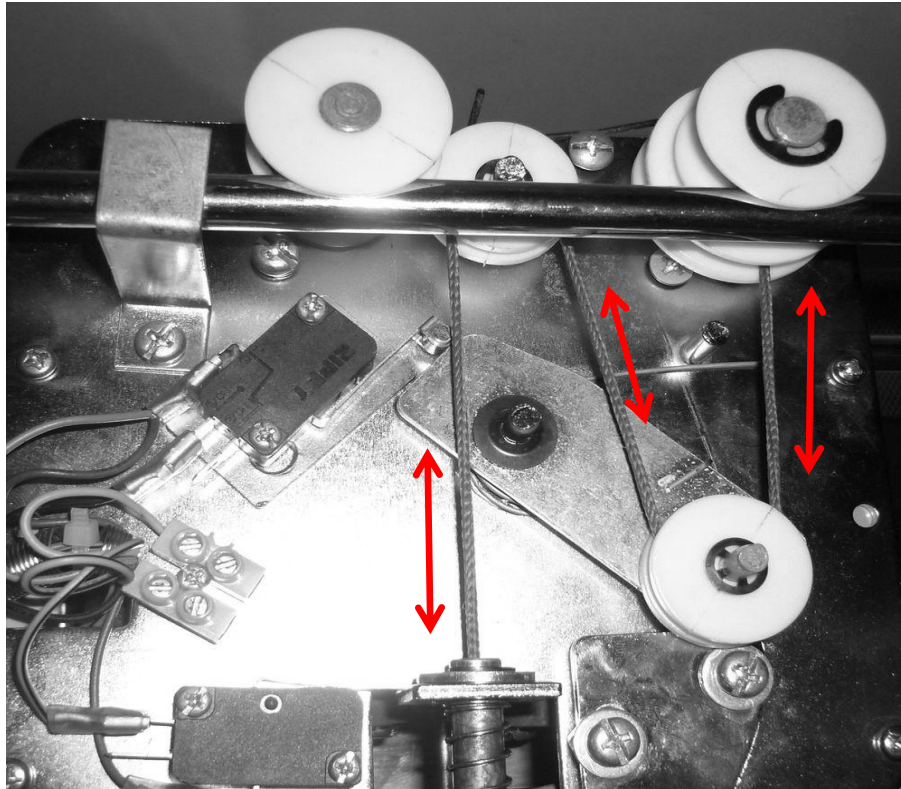
12 VDC PSU
EA1003



Prize Sensor (Viewed inside prize chute)



ROPE WINDING FOR THE CLAW



*** NOTE! ***

Rope length between notches is 1.40 meters (55") maximum.

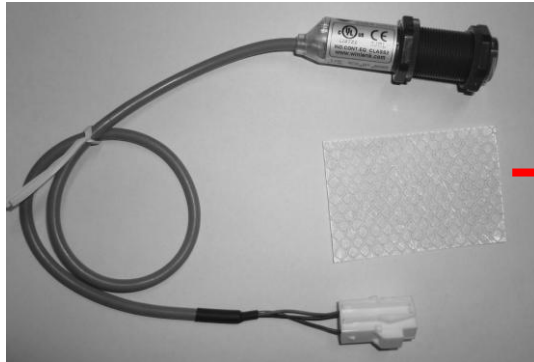
DISPLAY ACCESS,



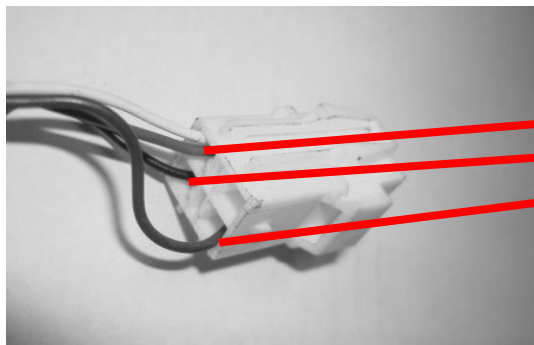
Unscrew the 4 screws of the display to give access as shown above



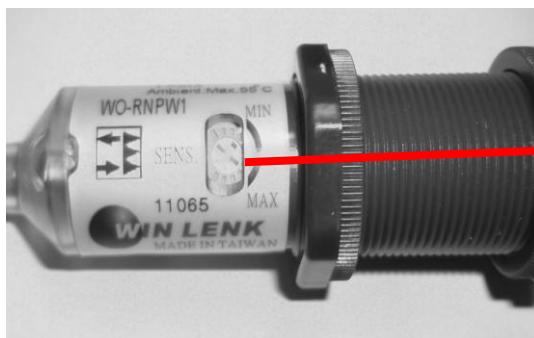
PRIZE SENSOR



Prize Sensor WO-RNPW1
Win Lenk Retro Reflective



Blue/White = GND
Black = Output Logic
Brown = 12 VDC



Sensitivity Adjustment



PARTS DESCRIPTION

■ COIN MECHANISMS

The coin mechanisms can be accessed inside the coin door to the right on the front of the machine cabinet.

■ CASH BOX

The cash box is located inside behind the coin door on the front of the machine cabinet.

■ SPEAKERS

Two speakers are located to the front of the cabinet below the control panel. Access is through the back of front door by unscrewing the metal cover.

■ CONTROL PANEL

Control Panel located in the center of the machine cabinet. The control panel can be accessed through the front door from the back by unscrewing the metal bracket.

STACK BUTTON: The Stack button is the Large Red round illuminated button. This button is used to start the game, to stop and Stack the blocks or to lower the claw to grab a prize and for test and program adjustments.

JOYSTICK: The joystick is use for moving the claw into the right position above a prize if a player wins

■ SERVICE CONTROLS:

The service panel is located above the cash box and accessed through the front door.

SERVICE BUTTON: Used to input credits to the game without activating the coin counter, and to perform test procedures in combination with the test button

TEST BUTTON: Used to perform the test mode, in combination with the Service button.

VOLUME KNOB: Used to adjust the speaker's sound level.

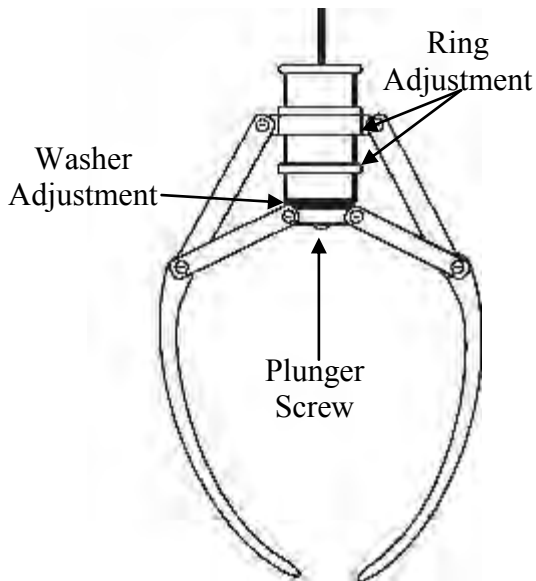


Test Button
On the Door



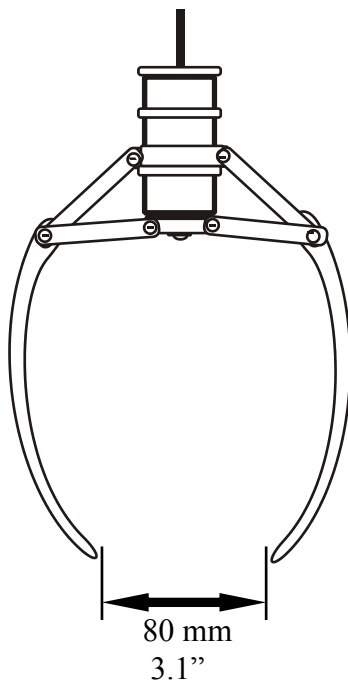


CLAW ADJUSTMENT

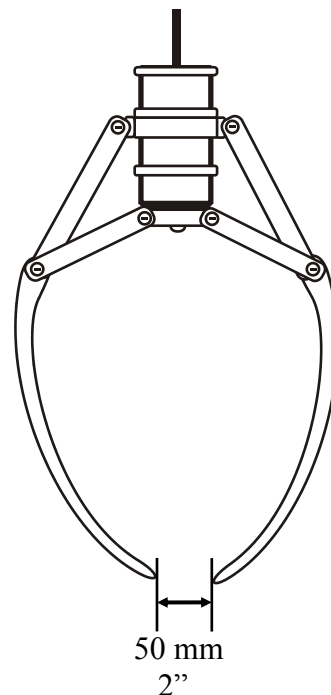


1. Mechanical Adjustments
 - **Distance between claw arms in the closed position** is adjusted by moving the two rings up and down the body of the claw. Close the claw manually and loosen ring grub screws. Adjust the position of the rings for the desired claw arm distance and tighten ring grub screws.
 - **Distance between claw arms in the Open position** is adjusted by moving the square washer at the base of the plunger. Loosen plunger screw and rotate the square washer to allow the claw to open wide or narrow.

Big toys claw adjustment



Smaller toys claw adjustment





Note: The claw will get hot during normal game operations



*** CAUTION! ***

Do not touch the Power Coil during the operation it may be extremely hot



LAMPS

*** WARNING! ***

Always turn **OFF** Mains power and unplug the game, before replacing any lamps.

Always allow time for cooling as Lamps that have been active for a time may still be too hot to touch.

■ COIN DOOR LAMPS

The coin door lamps all are 12V/DC LED or equivalent and can be accessed through the front door.

■ BUTTON LAMPS

The button lamps all are 12V/DC LED or equivalent and can be accessed through the front door.

■ HEADER LAMPS

There are standard FL fluorescent tubes for the Header Display. Access is by the removing of the machine header cover and accessing the tube and ballast from the front.

■ CABINET LAMPS

There are four 20 watt Energy Saver lamps in each top corner of the inside of the cabinet. Maximum four 25 watt Energy saver lamps can be use inside the cabinet.

*** CAUTION! ***

Always Replace the lamps with the same or equivalent size, wattage and voltage.



MAINTENANCE

CLEANING AND CHECK UP

■ EXTERIOR

Regularly dust and clean the external cabinet areas as required, using a soft water-damp cloth and mild soap. Check for blown bulbs and replace as required.

Any scratches or marks in the fiberglass or acrylic can be buffed out using car polish or cut and polish.

*** CAUTION! ***

Do not Use solvents on the panels as it may affect the artwork.

■ INTERIOR

Regularly dust and vacuum the interior of the cabinet, taking care to remove any objects that may have fallen on the PCBs. Check and tighten all fixing hardware and fasteners as required.

*** WARNING! ***

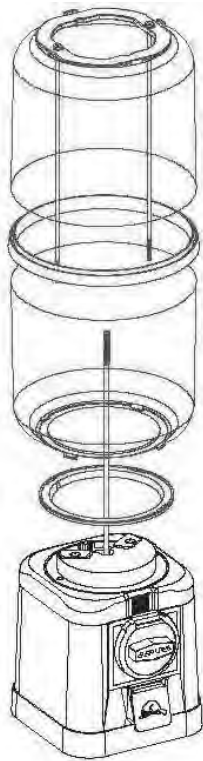
Always Turn **OFF** Mains power and unplugged the game, before cleaning the interior of the machine.

Always After cleaning the cabinet interior, check all harness connectors and restore all loose or interrupted connections.

Regularly check that all the display and button lamps are operating through the Sounds, Lamps and Display Test. Replace any lamps that are not operational.

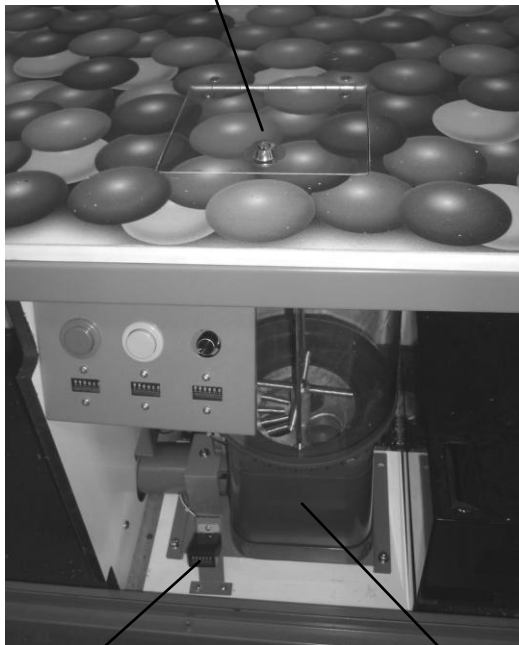


1" CAPSULE DISPENSER (*Optional*)

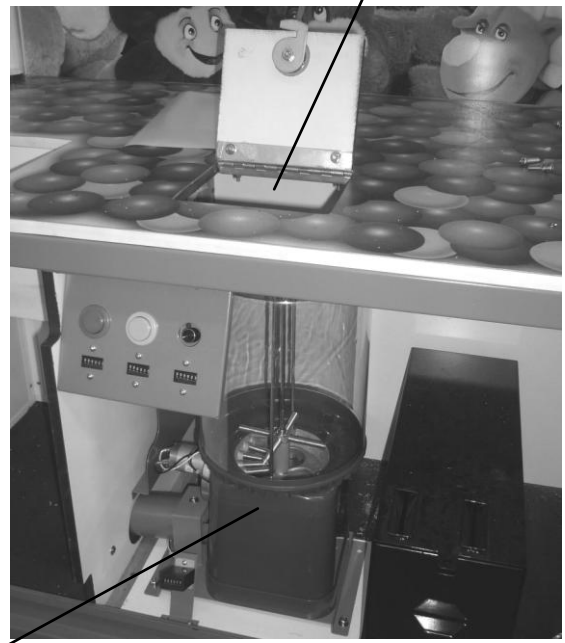


Capsule access door closed

Capsule access door opened



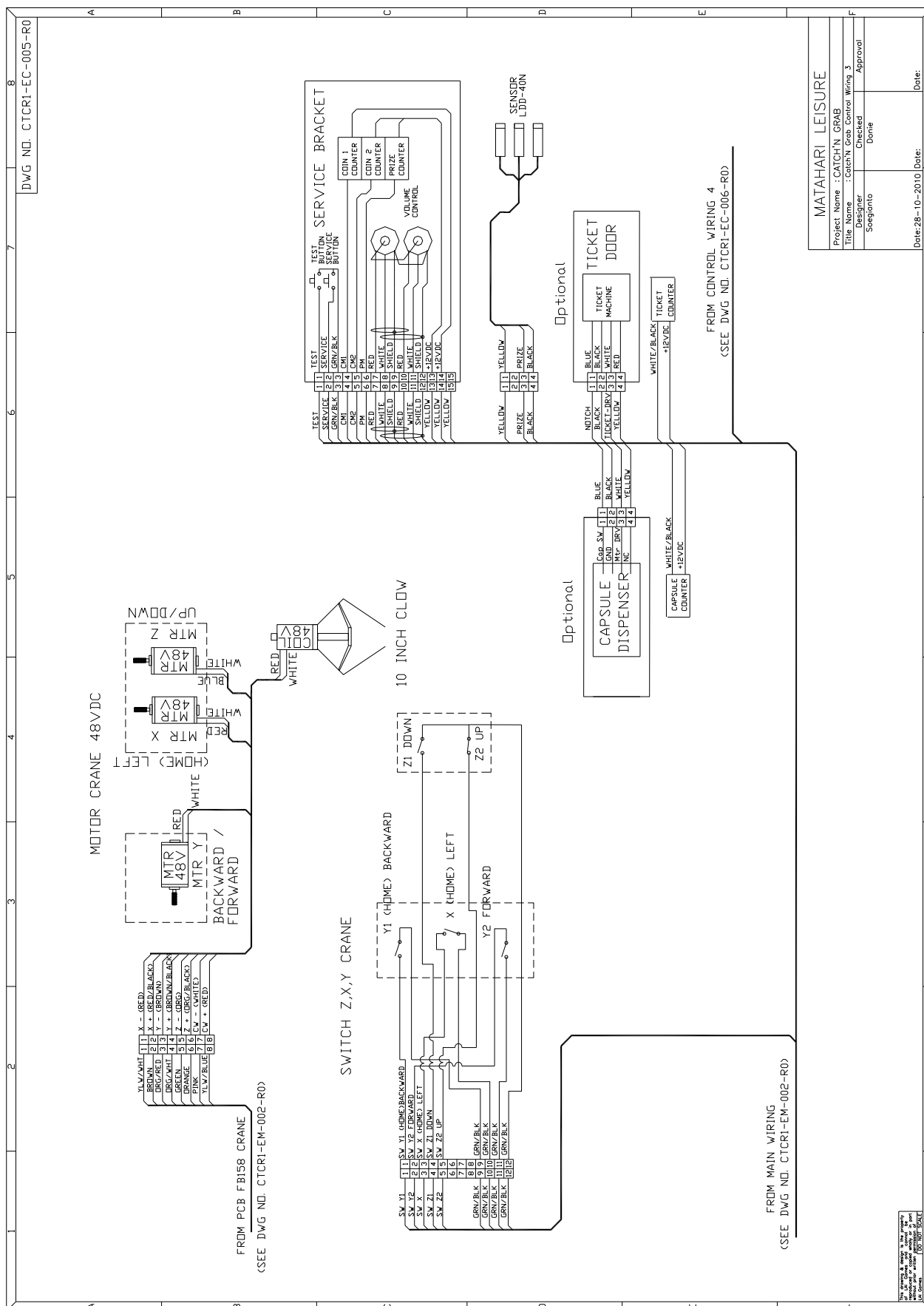
Capsule dispenser meter



1" Capsule dispenser



CAPSULE WIRING DIAGRAM (*Optional*)





FILLING IN THE CAPSULES (Optional)

Filling the 1" Capsule Dispenser,



With the Capsule Access door opened fill in the 1" capsule into the capsule dispenser as required

Filling the Capsule display tube,



Use Phillip Head Screw to unscrew the top cover of the tube



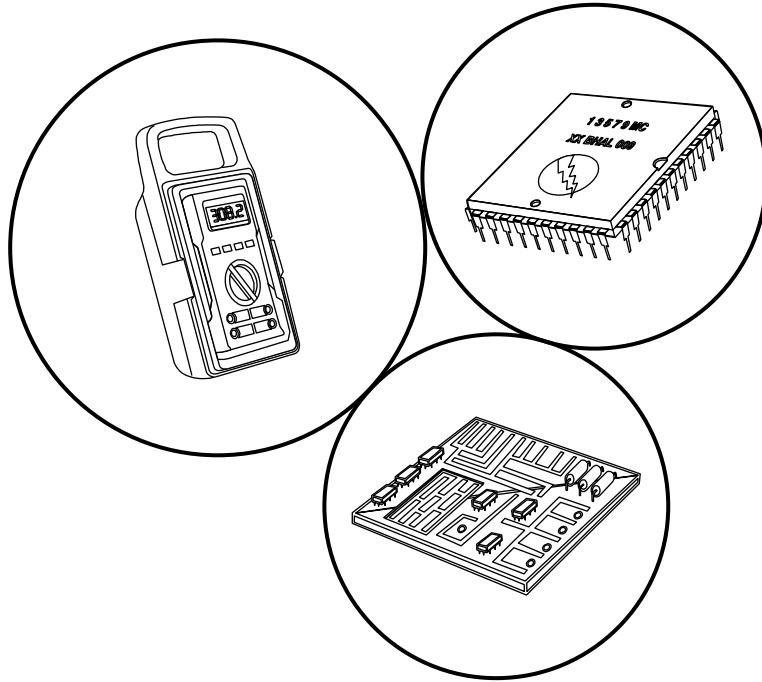
Lift the cover



Fill in the capsule inside the tube and replace in reverse order



SECTION B: TECHNICAL DETAILS



It is advised that anybody using SECTION B for repairing or modifying any of the components of the game should be a qualified technician, having at least a basic knowledge of digital components, integrated circuits and electricity.





CONNECTION DETAIL

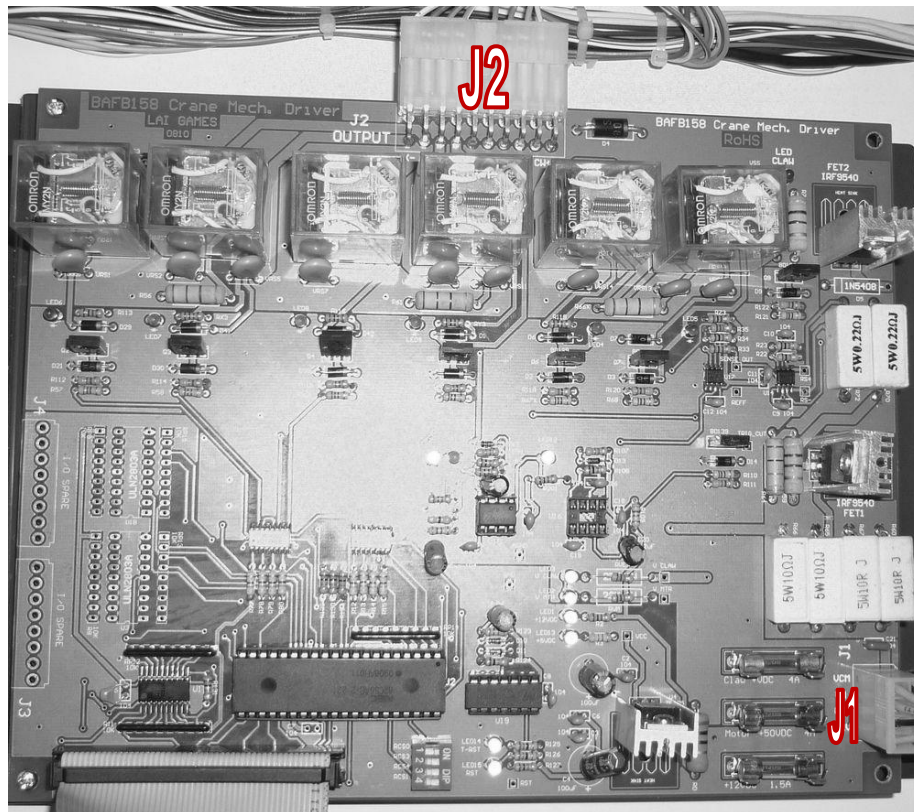
Main PCB 28 Way Edge Connector

Components Side	Solder side
GND 1	A GND
GND 2	B GND
+12VDC 3	C +12V
+12VDC 4	D +12V
- P1A7 5	E P1B7 Test button input
Z Down Switch 2 P1A6 6	F P1B6 Service button input
Z Up Switch 1 (home) P1A5 7	H P1B5 Start Button input
X Left Home Position switch P1A4 8	J P1B4 -
Right Joystick Input P1A3 9	K P1B3 Ticket notch input
Left Joystick Input P1A2 10	L P1B2 Coin 1 input
(Back)Up Joystick Input P1A1 11	M P1B1 Coin 2 input
(Front)Down Joystick Input P1A0 12	N P1B0 -
(watch dog) P1C7 13	P P2C7 Display 7 Segment data
- P1C6 14	R P2C6 Output
- P1C5 15	S P2C5 Select button light
P1C4 16	T P2C4 Display, Display clock
Prize Sensor input P1C0 17	U P2C0 Display data 1
Select Button input P1C1 18	V P2C1 Display data 2
Y Backward Switch 1 (home) P1C2 19	W P2C2 -
Y Forward Switch 2 (NC) P1C3 20	X P2C3
GND 21	Y GND
GND 22	Z GND

JST 7	JST 5
Light level 1 P2A0 1	1 P2B0 (PNP) Ticket Drive
Light level 2 P2A1 2	2 P2B1 Start Button Light Drive
Light level 3 P2A2 3	3 P2B2 Coin 1 Counter
Light level 4 P2A3 4	4 P2B3 Coin 2 Counter
Light level 5 P2A4 5	5 P2B4 Prize Counter Drive
Light Led Winner 6 P2A5 6	6 P2B5 Beacon /Led Light
Front Up Light 1 P2A6 7	7 P2B6 Crane Control
Front Down Light 2 P2A7 8	8 P2B7 Capsule Counter

Note: All switches that not write (NC) are NO (Normally Open).

FB158 PCB Relay Connector detail,



J2 (OUTPUT) 90 degree connector

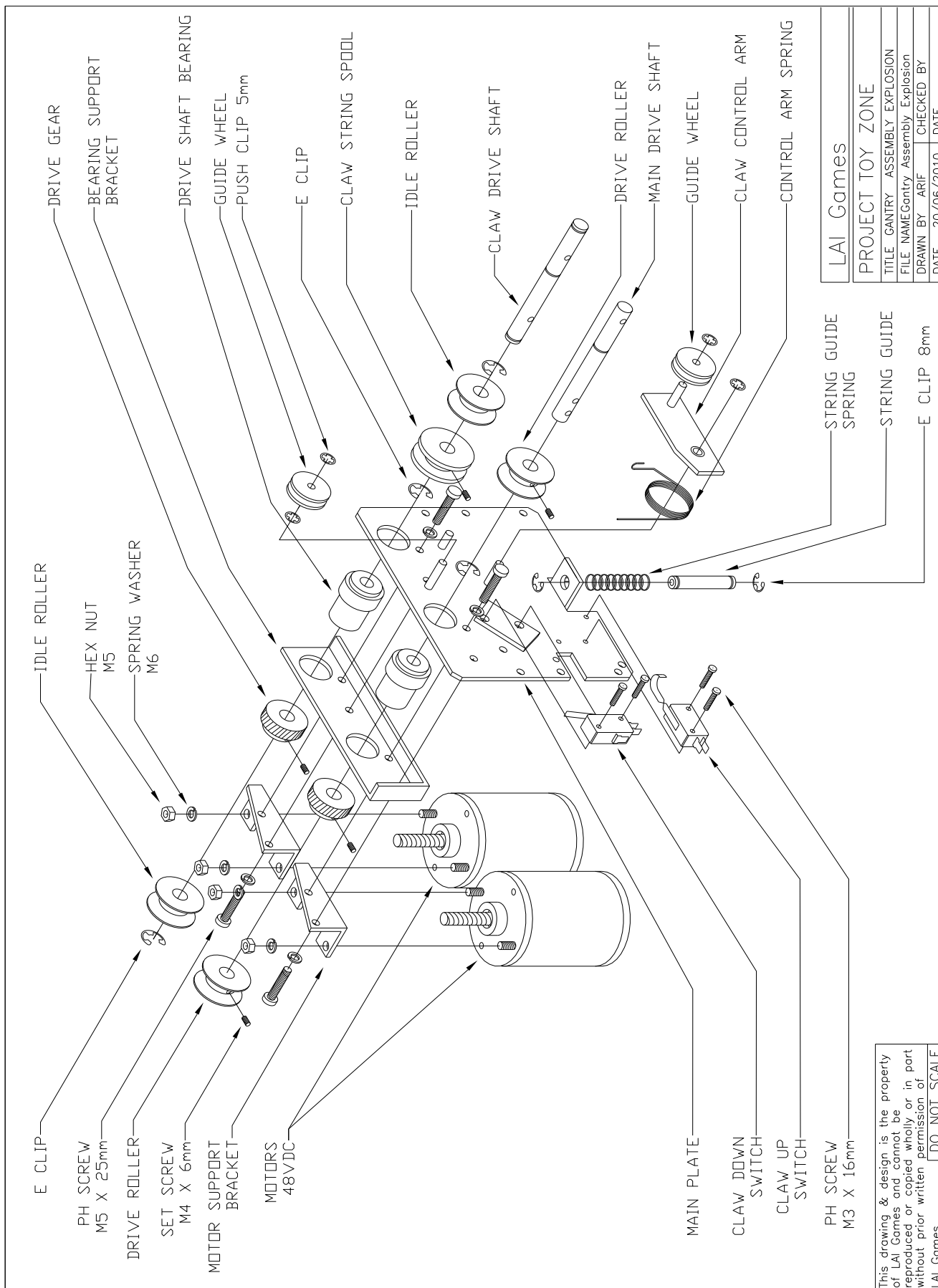
Pin 11 = Left direction (X -)
Pin 12 = Right direction (X +)
Pin 13 = Backward direction (Y -)
Pin 14 = Forward Direction (Y +)
Pin 15 =
Pin 16 = Down Direction (Z -)
Pin 17 = Up Direction (Z +)
Pin 18 = Claw Active (CW +)
Pin 19 = Claw In active (CW -)
Pin 20 =

J1 (Power Connector)

Pin 1 = +12VDC
Pin 2 = GND
Pin 3 = +48VDC
Pin 4 = +48VDC
Pin 5 = GND 12V DC
Pin 6 = -
Pin 7 = -
Pin 8 = GND 48V DC

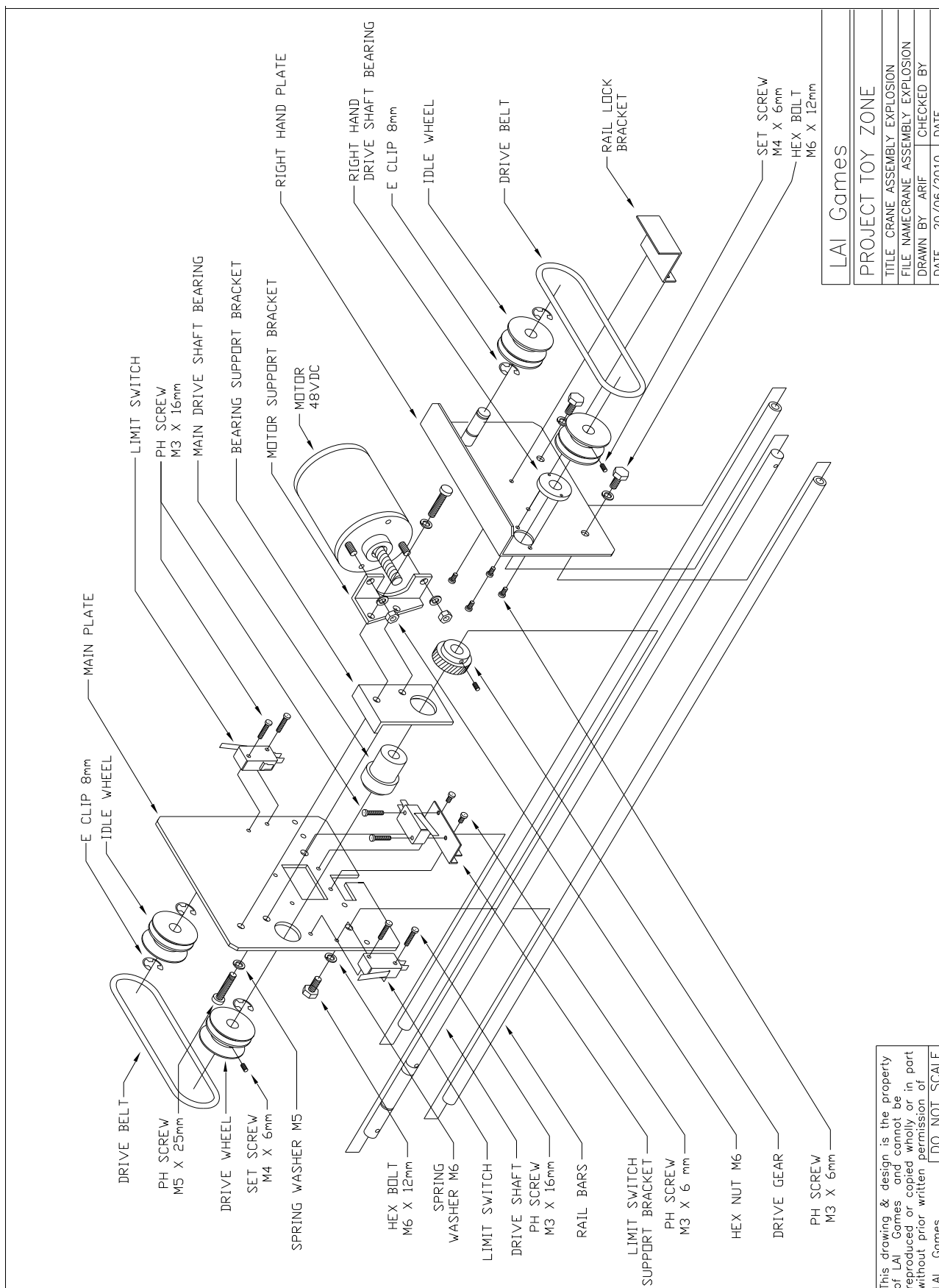


GANTRY EXPLODED VIEW





CRANE EXPLODED VIEW

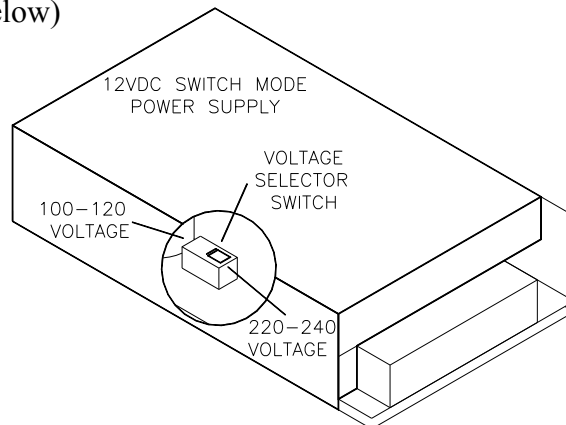




MAINS VOLTAGE ADJUSTMENT

■ POWER SUPPLY

The Switch Mode Power Supply has a switch to set the mains voltage range. It is located at the rear of the game cabinet, and is accessed via the Service door. Use a thin blade screwdriver to move the selector switch to the desired mains voltage (See Diagram Below)

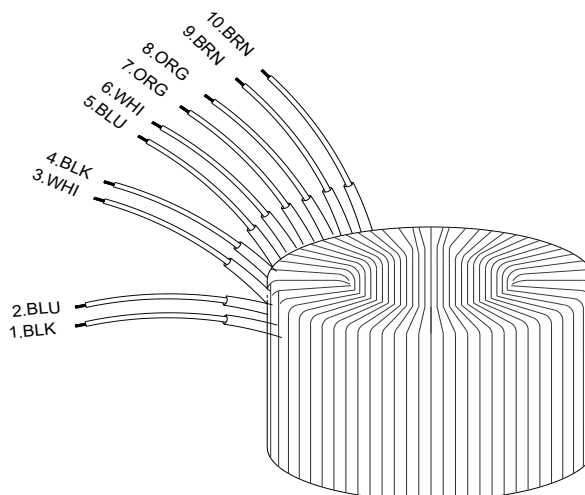
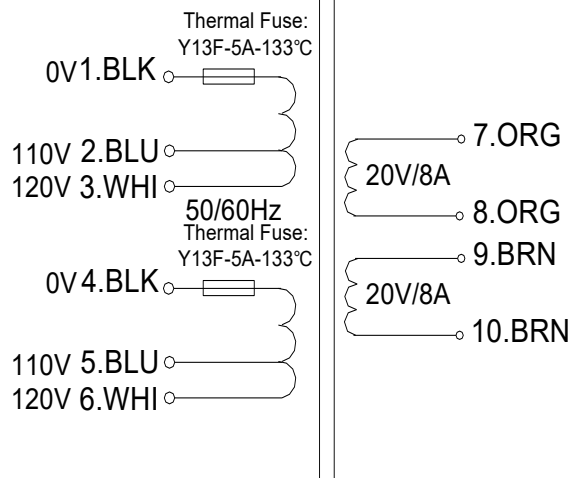


■ TRANSFORMER CONNECTORS

Locate the machine transformer(s) in the base of the cabinet. If unsure of the location of the transformer(s), refer to Parts location diagram page of this manual.

6 WAY CONNECTOR PINOUT

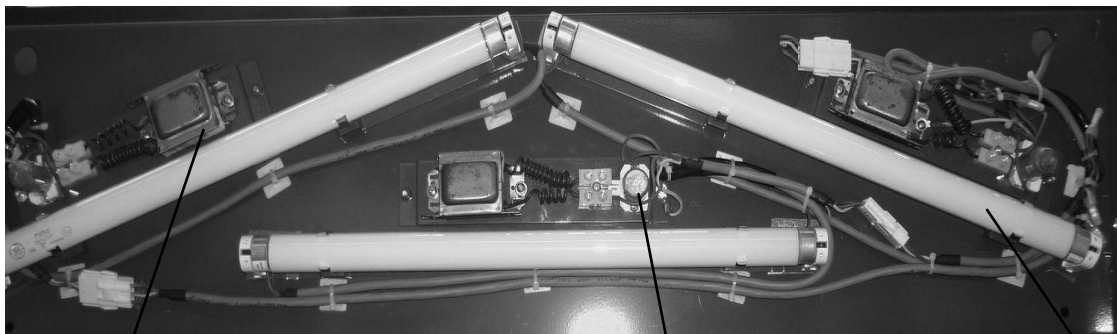
PIN	FUNCTION
1	0 VAC
2	110 VAC
3	120 VAC
4	0 VAC
5	110 VAC
6	120 VAC
7	0 VAC
8	20 VAC
9	0 VAC
10	20 VAC





■ FLUORESCENT TUBE BALLASTS AND STARTERS

Locate the florescent tube ballasts and starters on the top of the cabinet behind the Game Header. If unsure of the location of any ballasts or starters, refer to Parts location diagram page of this manual. These have to be removed and replaced with an equivalent wattage as your local mains voltage level.

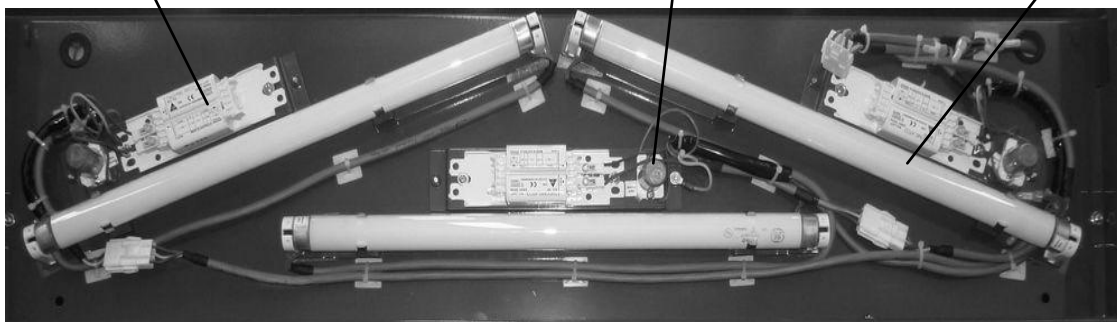


**110 VAC
Ballast**

Ballast

Starter

**18 Watt
Fluorescent**



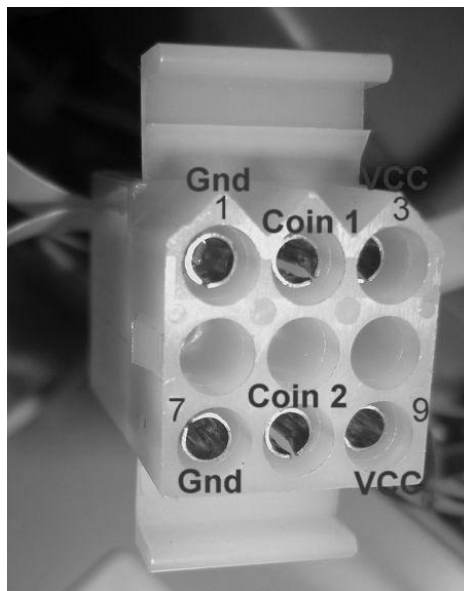
**220 VAC
Ballast**



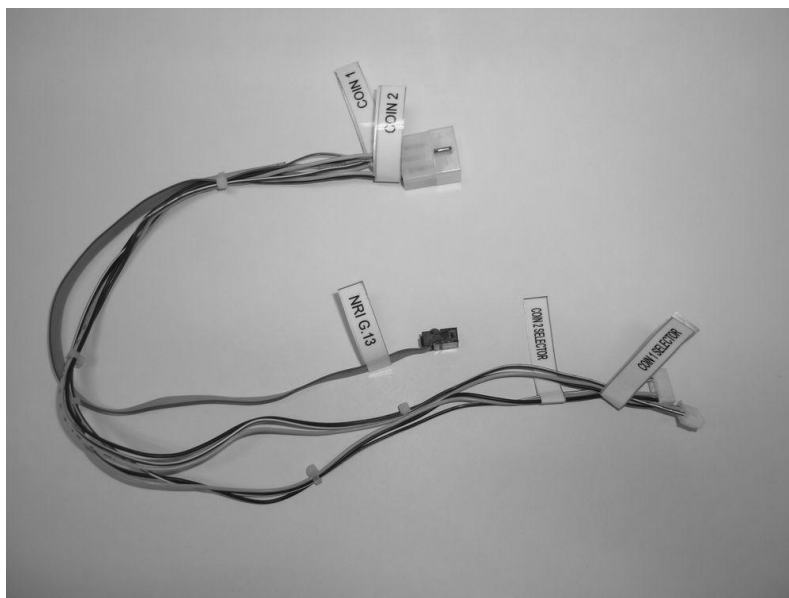
COIN OPTIONS REFERENCE GUIDE

LAI Games have installed on the coin door is a 9 way Molex connector. This connector can be used for connection to most electronic coin systems and electronic coin comparators.

Please see below picture to understand the connection setup,



The Coin harness supplied with the game,

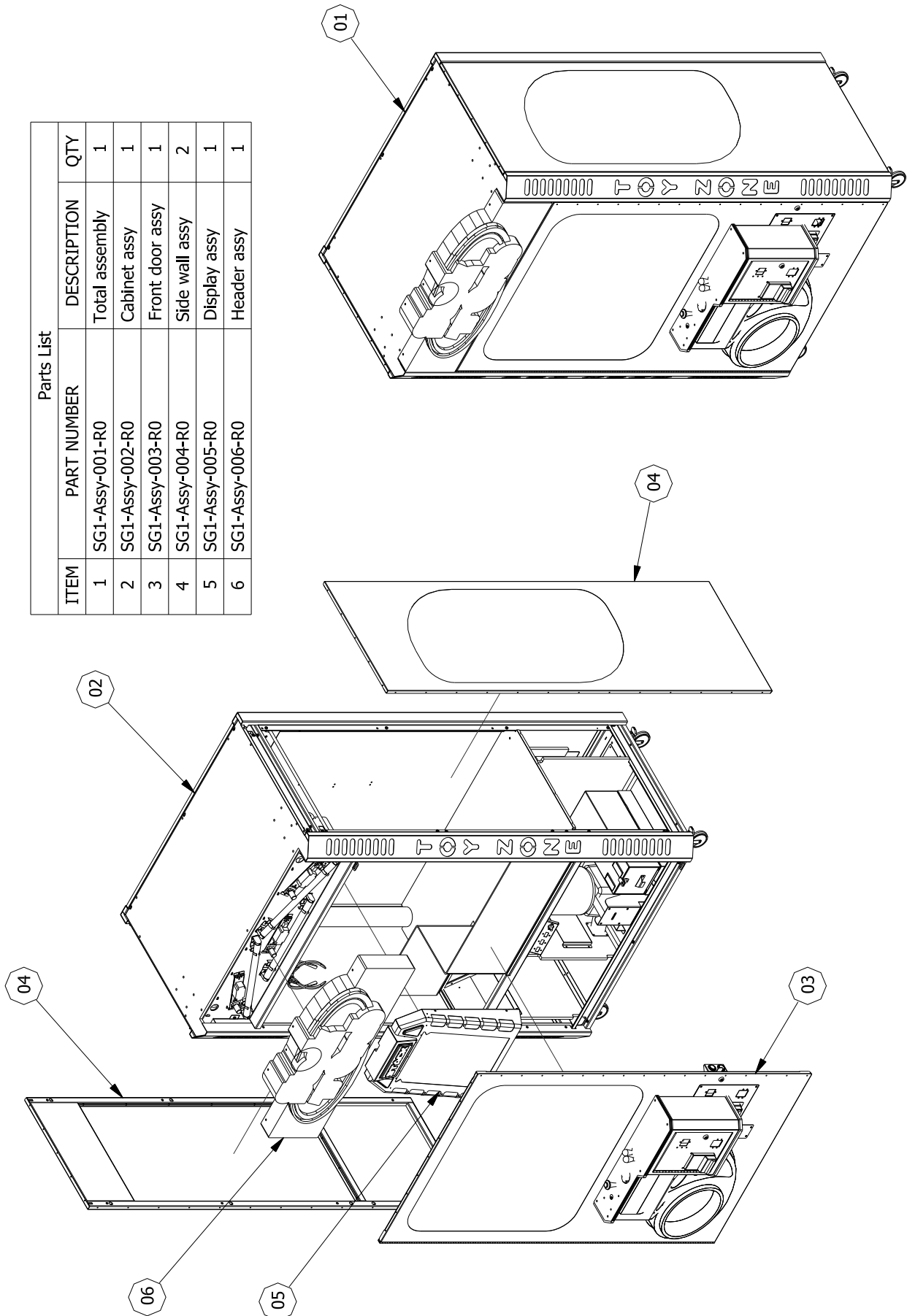


The Game is supplied standard with a harness to fit to NRI G13 and 2x Electronic Coin Comparator (**LAI Games** standard option). Other types of coin electronic harnessing can be supplied as requested, please contact your nearest **LAI Games** distributor.



3 D PART EXPLODE VIEW

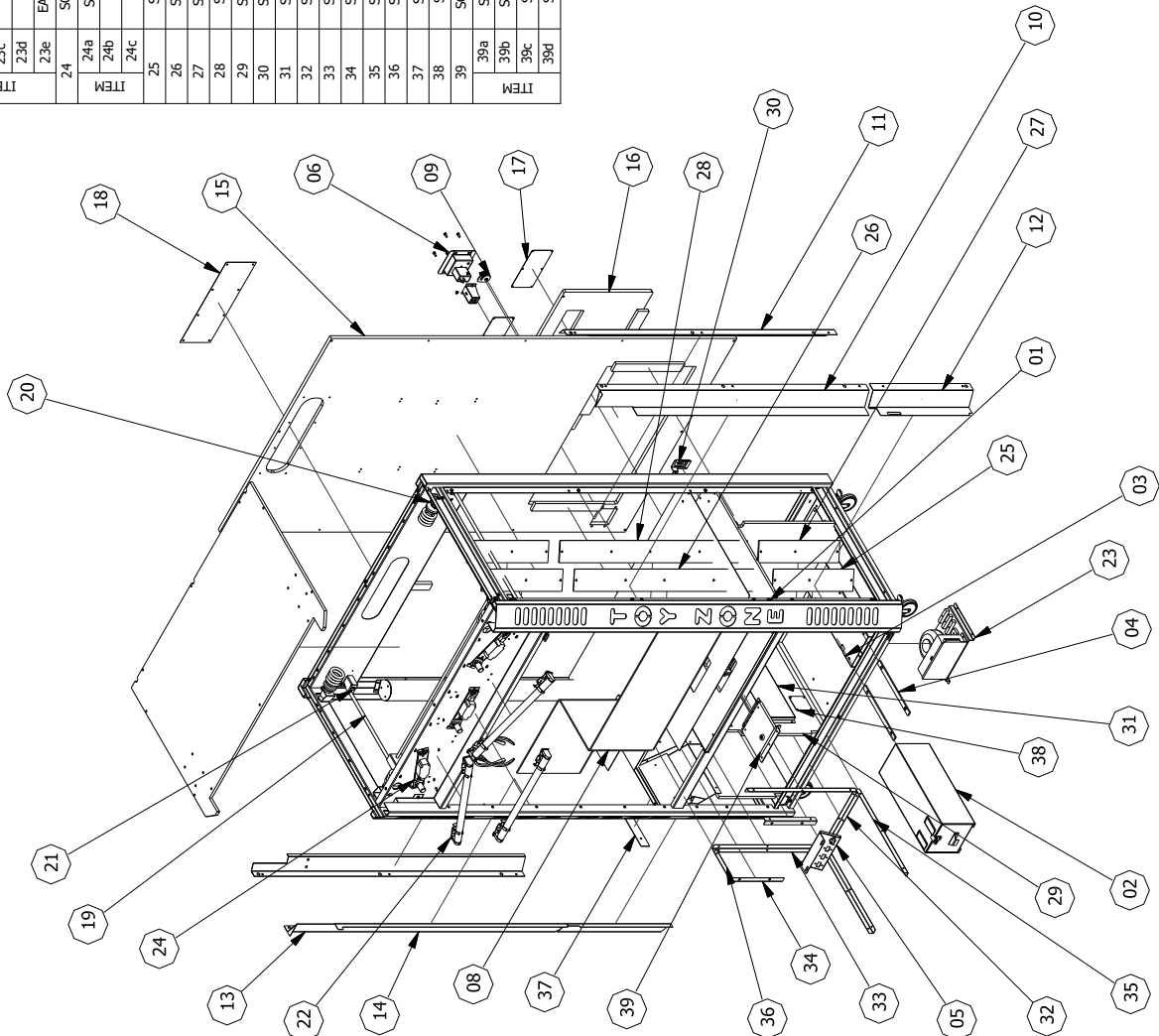
Parts List			
ITEM	PART NUMBER	DESCRIPTION	QTY
1	SG1-Assy-001-R0	Total assembly	1
2	SG1-Assy-002-R0	Cabinet assy	1
3	SG1-Assy-003-R0	Front door assy	1
4	SG1-Assy-004-R0	Side wall assy	2
5	SG1-Assy-005-R0	Display assy	1
6	SG1-Assy-006-R0	Header assy	1





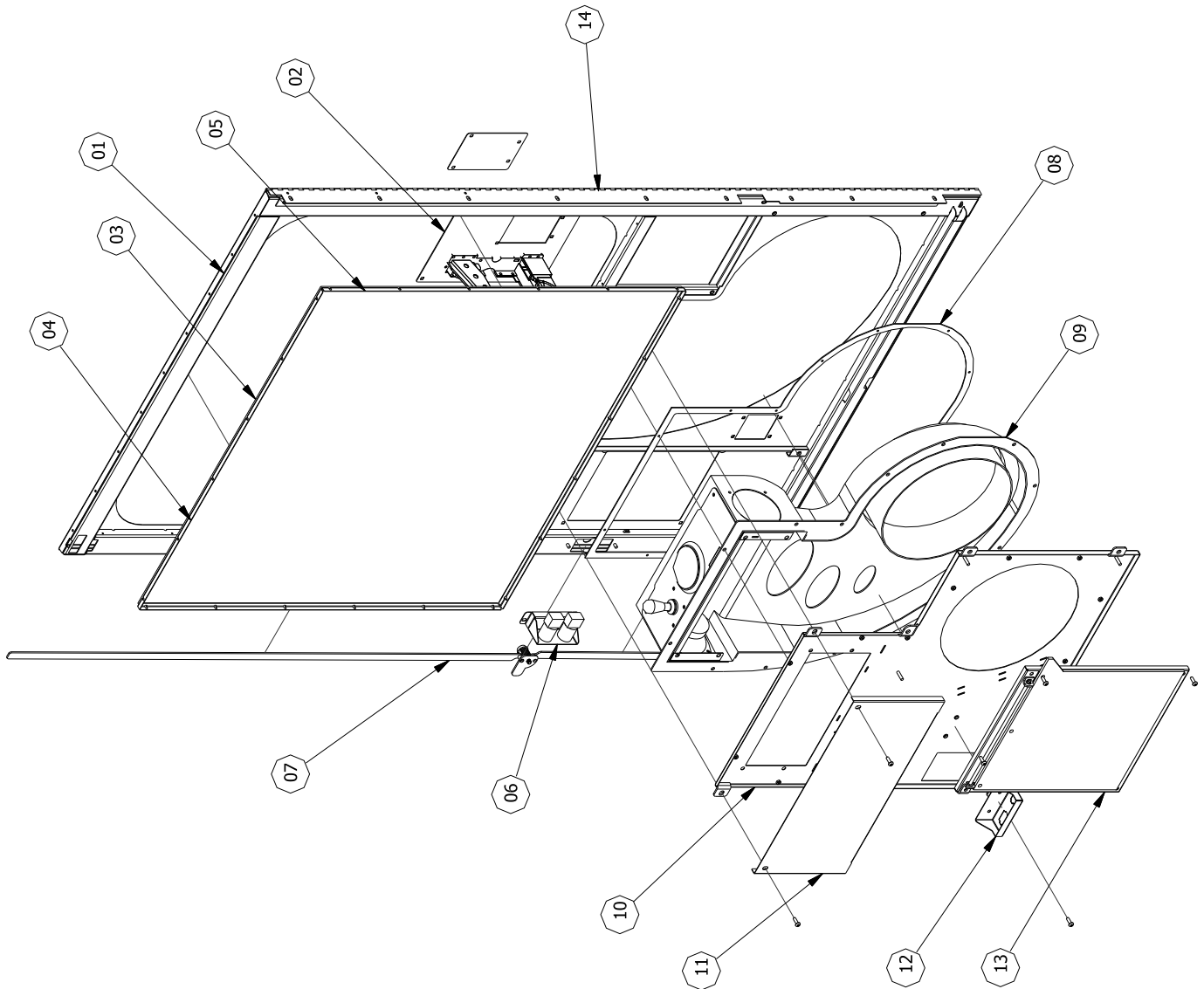
NO	PART NUMBER	DESCRIPTION	QTY
ITEM	01	SGI-SA-007-R0	Frame Assy
	01a	SGI-SA-001-R2	Frame SA
	01b	SGI-FW-001-R3	Base
	01c	HM 0091	Caster wheel 3"
	01d	SGI-FW-006-R3	Prize hopper 2
ITEM	01e	SGI-FW-009-R0	Prize hopper base
	01f	SGI-FW-007-R1	Prize hopper 1
	01g	SGI-FW-008-R2	Prize door stopper
	01h	SGI-FW-003-R2	Prize base
	01i	SGI-FP-015-R1	Prize base cover
	01k	SGI-FW-052-R0	Base vent
	01l	SGI-FW-002-R2	Top wall
	02	SGI-Assy-008-R0	Cash box Assy
	02a	SGI-SA-010-R0	Cash box cover SA
	02b	SGI-SA-011-R0	Housing dash box SA
ITEM	03	SGI-FM-023-R0	Cash lifter
	04	SGI-FP-009-R0	Cash box base
	05	SGI-Assy-009-R0	Service bracket Assy
	05a	SGI-SA-004-R2	Service bracket SA
	05b	EA1252	COIN COINER 12V GEAR MOUNTING
ITEM	05c	EO0689	Potensio carbon
	05d	EP0602	Knob volume
	05e	EA0520	SWITCH SMALL ROUND GREEN BUTTON
	05f	EA0519	SWITCH SMALL ROUND GREEN BUTTON
	06	SGI-Assy-010-R0	DB box Assy
ITEM	06a	SGI-FM-048-R0	DB box front
	06b	SGI-FM-049-R0	DB box housing
	06c	EA1356	BINDING POST
	06d	EA1358	SAIT GREEN FILTER FOR CE MACHINE
	06e	EA0649	IEC TYPE NOISE EMI FILTER
	06f	MS1 H001	DB BOX WARRIES
	06g	EA0635	POWER LED INCLUDED DEC TO 3 PIN USA
	06h	EA0636	POWER LED INCLUDED DEC TO 2 PIN IND
	06i	EA0637	POWER LED INCLUDED DEC TO 3 PIN AU
	06j	EA0639	POWER LED INCLUDED DEC TO 3 PIN UK
ITEM	07	SGI-FM-019-R4	Prize chute
	08	SGI-FP-005-R2	Plastic guard
	09	SGI-FM-020-R1	Lock cam 1
	10	SGI-FM-017-R2	Front LED cover top right
	11	SGI-FM-018-R1	Back LED cover top
	12	SGI-FM-015-R0	Front LED cover lower right
	13	SGI-FM-016-R2	Front LED cover top left
	14	SGI-FM-022-R0	Front LED cover lower left
	15	SGI-FW-004-R1	Back wall
	16	SGI-FW-005-R0	Back door
ITEM	17	SGI-FM-036-R0	Back door vent
	18	SGI-FM-033-R0	Back wall vent
	19	SGI-SA-008-R1	Ball SA
	20	SGI-Assy-011-R0	Neon bracket 2 Assy
	20a	SGI-FM-030-R0	Neon bracket 2
	20b	EP 0435	LAAMP FITTING HOLDER E26 UL
	20c	EP 0280	24W Tornado Daylight
	21	SGI-FM-050-R0	Bumper bracket
	22	SGI-Assy-012-R0	Neon holder 2 Assy
	22a	SGI-FM-029-R0	Neon holder 2
ITEM	22b	EA 0205	Neon 1.5W Cool white
	22c	EP 0434	END CAP HOLDER MODEL 7134H
	22d	HM 0021	Neon dip
			6 Pcs

NO	PART NUMBER	DESCRIPTION	QTY
ITEM	23	SGI-Assy-023-R0	Power supply bracket Assy
	23a	SGI-FM-036-R1	Power supply bracket
	23b	EA 0832	Two Pin 3.15V 150mA 0.500 380A
	23c	EA 1003	Switching 120V 12.5A S-190-12
	23d	EA 2422	Diode 1N 5400
ITEM	23e	EA 0040 ; EA 0039	Four 7A Fuse 4A
	24	SGI-Assy-024-R0	Balast bracket Assy
	24a	SGI-FM-028-R0	Balast bracket
	24b	EA 0326	Balast CE 40W 1820W ATCO
	24c	EA 0304	Neon Starter 2U 220-240V
ITEM	25	SGI-FP-001-R0	LED plastic cover
	26	SGI-FP-002-R0	LED plastic cover middle
	27	SGI-FP-003-R0	LED plastic cover 2
	28	SGI-FP-004-R0	LED plastic cover middle 2
	29	SGI-FM-051-R0	PCB base
	30	SGI-FM-027-R1	Controller bracket
	31	SGI-FM-056-R0	Manual book bracket
	32	SGI-FM-058-R0	Rainforce plate 1
	33	SGI-FM-059-R0	Rainforce plate 2
	34	SGI-FM-060-R0	Rainforce plate 3
ITEM	35	SGI-FM-057-R0	Rainforce plate 4
	36	SGI-FM-074-R0	Rainforce plate 5
	37	SGI-FM-072-R0	Reflector bracket
	38	SGI-FM-071-R0	Capsule dispenser cover
	39	SGI-Assy-025-R0	Dispenser cover Assy
	39a	SGI-FW-011-R0	Dispenser door
	39b	SGI-FW-064-R1	Dispenser door stopper
	39c	SGI-M-002-R2	Dispenser hinge
	39d	SGI-FM-20-R2	Lock cam 1
			1 Pcs



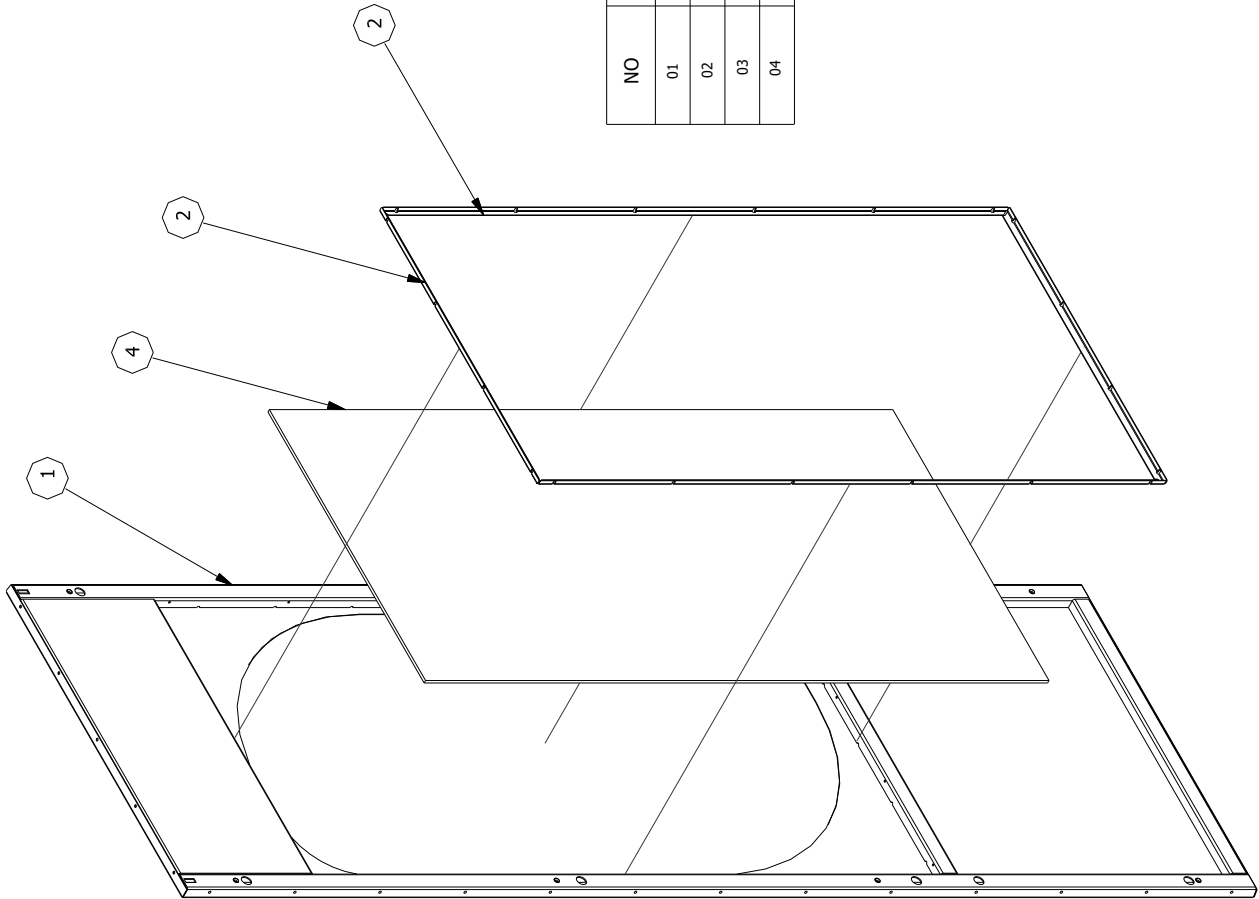


NO	PART NUMBER	DESCRIPTION	QTY
01	SG1-SA-005-R3	Front door SA	1 Set
02	SG1-Assy-026-R0	Coin & DBA assy	1 Set
ITEM	02a	SG1-FM-031-R0	Coin & DBA bracket
	02b	HW00014	Coin mechanism
	02c	SG1-FM-054-R0	DBA cover
03	SG1-FG-001-R0	Front door glass	1 Pcs
04	SG1-FM-045-R1	Front glass clamp 2	2 Pcs
05	SG1-FM-044-R0	Front glass clamp 1	2 Pcs
06	SG1-Assy-027-R0	Lock cam cover assy	1 Set
ITEM	06a	SG1-FM-035-R2	Lock cam cover
	06b	EA0520	Switch Small Round Green Button P/N: H5-02 M5
	06c	EA0519	Switch Small Round Red Button P/N: H5-02 M5
07	SG1-Assy-028-R0	Front door lock assy	1 Set
ITEM	07a	SG1-FM-021-R0	Lock cam 2
	07b	SG1-FM-042-R1	Upper rod key
	07c	SG1-FM-043-R0	Lower rod key
	07d	Rivet nut M5	Spacer
08	SG1-FM-047-R0	Front panel bracket	1 Pcs
09	SG1-Assy-029-R0	Control panel aassy	1 Pcs
ITEM	09a	SG1-FP-008-R1	Control panel
	09b	SG1-FM-041-R2	Control panel bracket
	09c	SG1-FM-037-R1	Speaket cover
	09d	SG1-FP-014-R2	Joystick base
	09e	EA 1206	Speaker 4"
	09f	EA 0591 A	Joystick
	09g	EA 0523 B	Red button big
	09h	SG1-FM-073-R1	Joystick support
	10	SG1-FM-012-R5	Front panel clamp SA
11	SG1-FM-34-R1	Front door cover	1 Pcs
12	SG1-FM-55-R0	Connector bracket	1 Pcs
13	SG1-Assy-30-R0	Prize door base assy	1 Set
ITEM	13a	SG1-SA-007-R1	Prize door base SA
	13b	SG1-FM-046-R0	Prize door guide 2
	13c	SG1-SA-006-R1	Prize door SA
	13d	SG1-FP-007-R1	Prize door
	13e	EP 0636	Bushing Ø6
14	SG1-M-001-R2	Front door hinge	1 Pcs



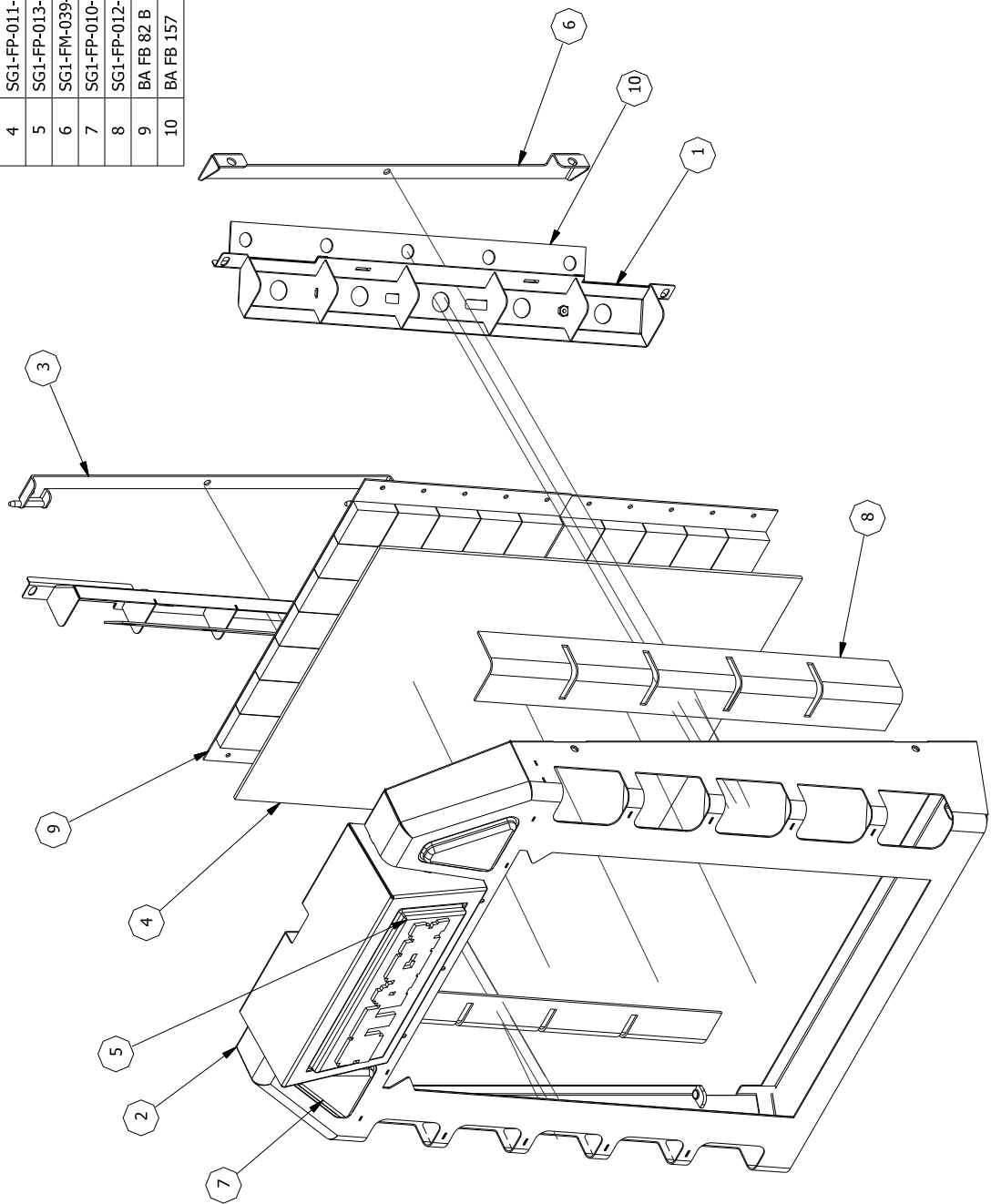


NO	PART NUMBER	DESCRIPTION	QTY
01	SG1-SA-002-R1	Side wall SA	1 Set
02	SG1-FM-024-R0	Side glass clamp 1	2 Pcs
03	SG1-FM-025-R0	Side glass clamp 2	2 Pcs
04	SG1-FG-002-R0	Side glass	2 Pcs



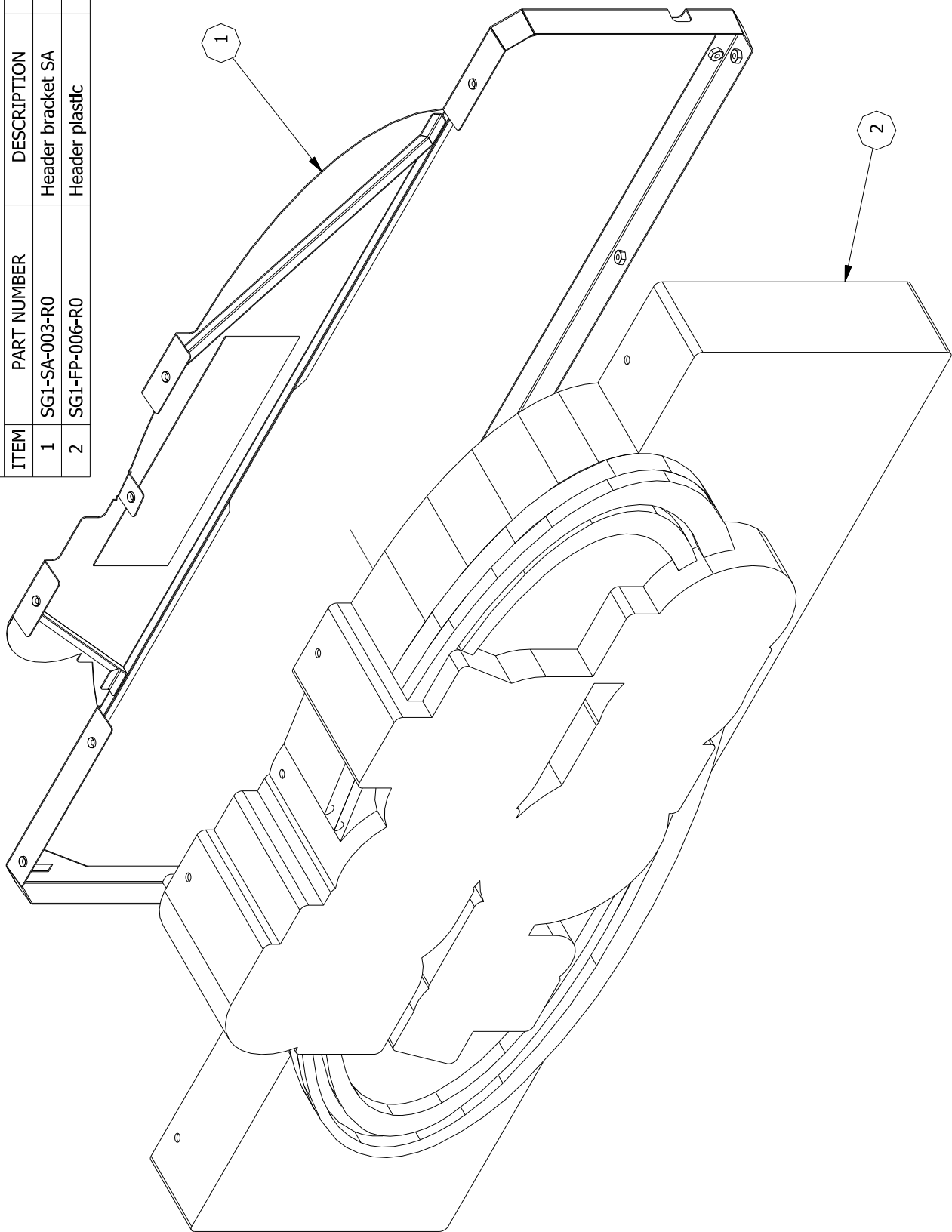


Parts List			
ITEM	PART NUMBER	DESCRIPTION	QTY
1	SG1-SA-013-R2	Lamp cover bracket	2
2	SG1-SA-009-R2	Display SA	1
3	SG1-SA-014-R1	Display hook SA	1
4	SG1-FP-011-R1	Display acrylic	1
5	SG1-FP-013-R1	Winner mold	1
6	SG1-FM-039-R1	Display hook 2	1
7	SG1-FP-010-R0	Side mold display	2
8	SG1-FP-012-R0	Prismatic cover	2
9	BA FB 82 B	FB82A	2
10	BA FB 157	PCB lamp	2



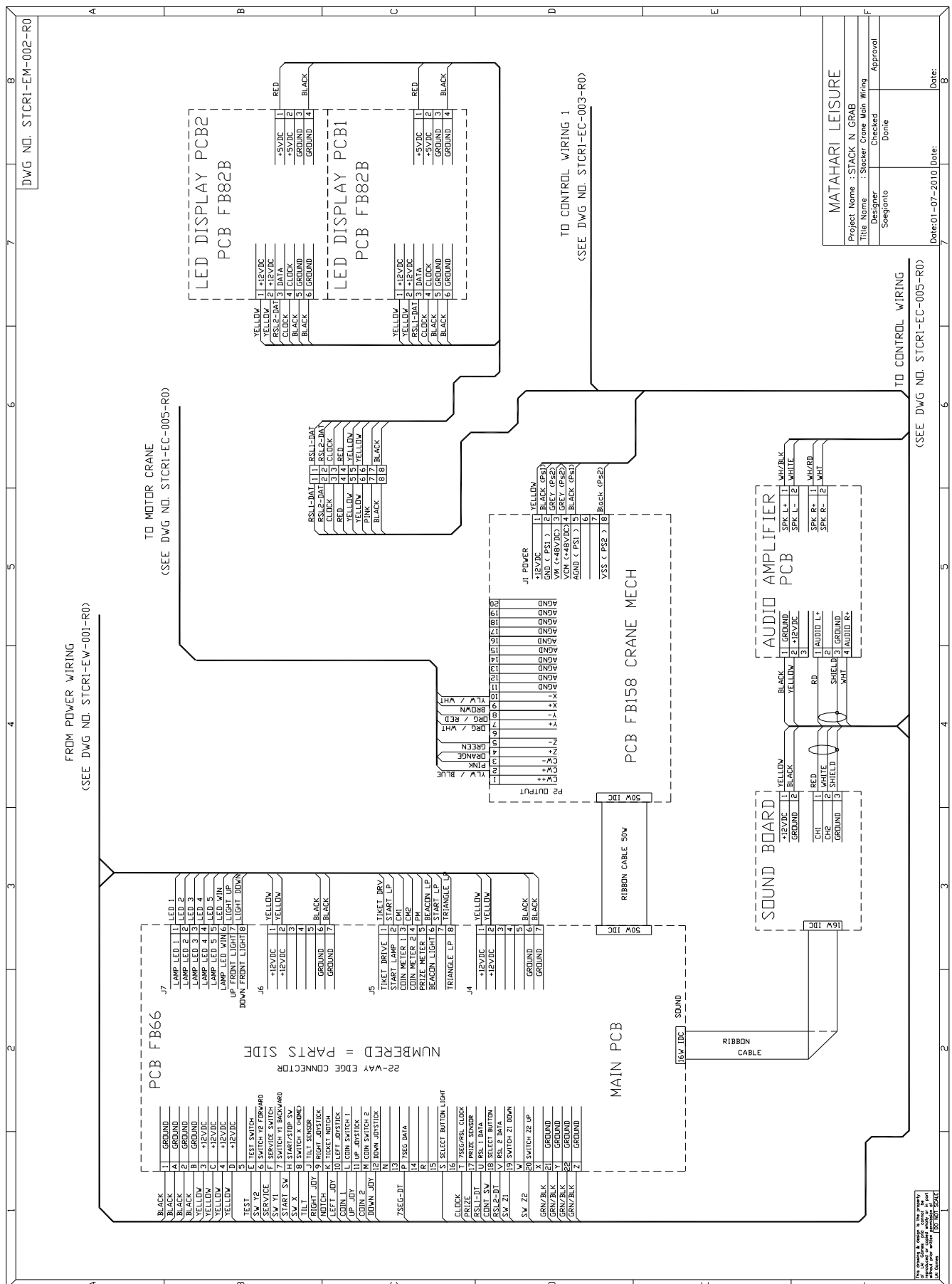


Parts List			
ITEM	PART NUMBER	DESCRIPTION	QTY
1	SG1-SA-003-R0	Header bracket SA	1
2	SG1-FP-006-R0	Header plastic	1



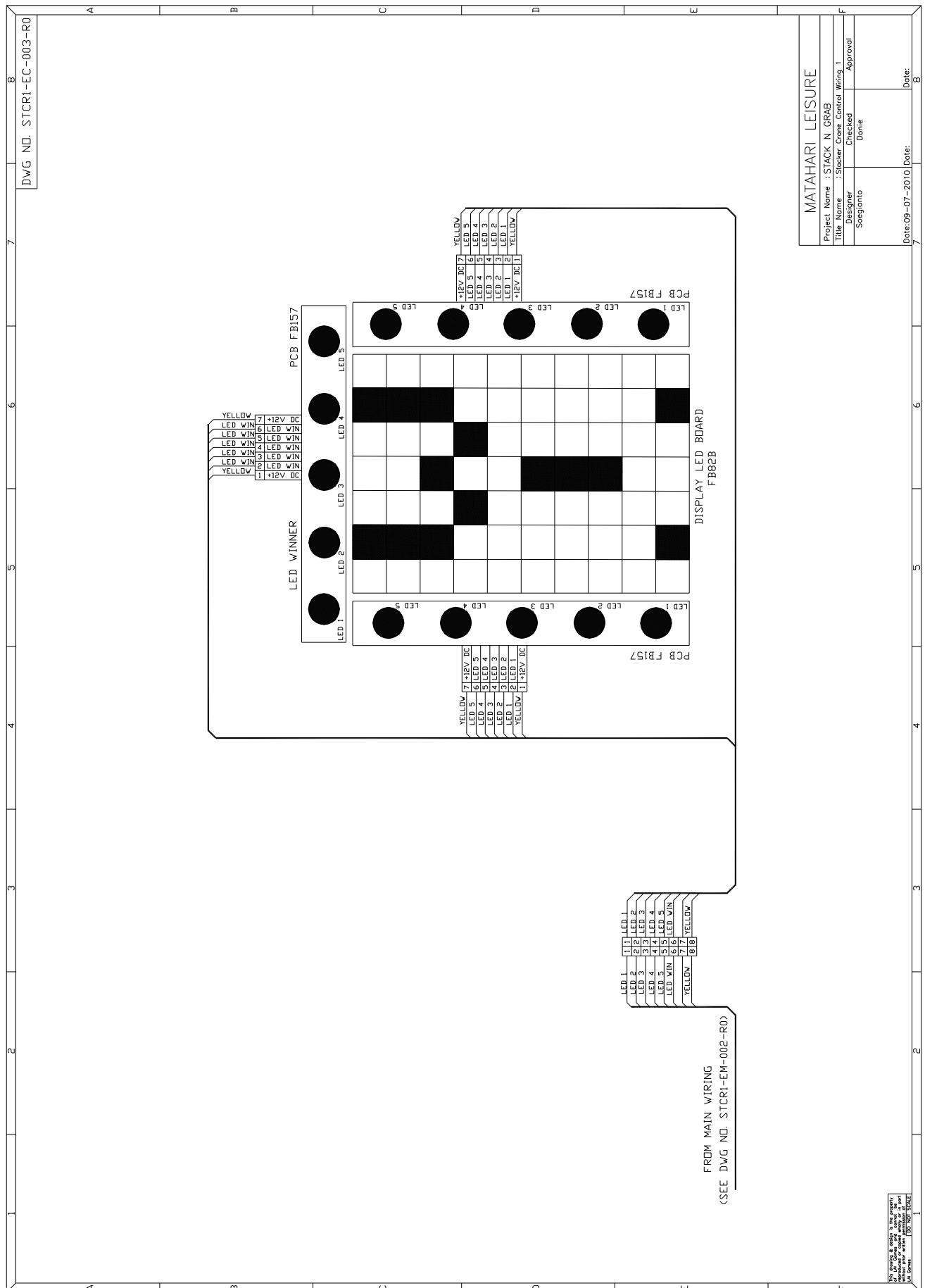


STACK 'N GRAB MAIN WIRING





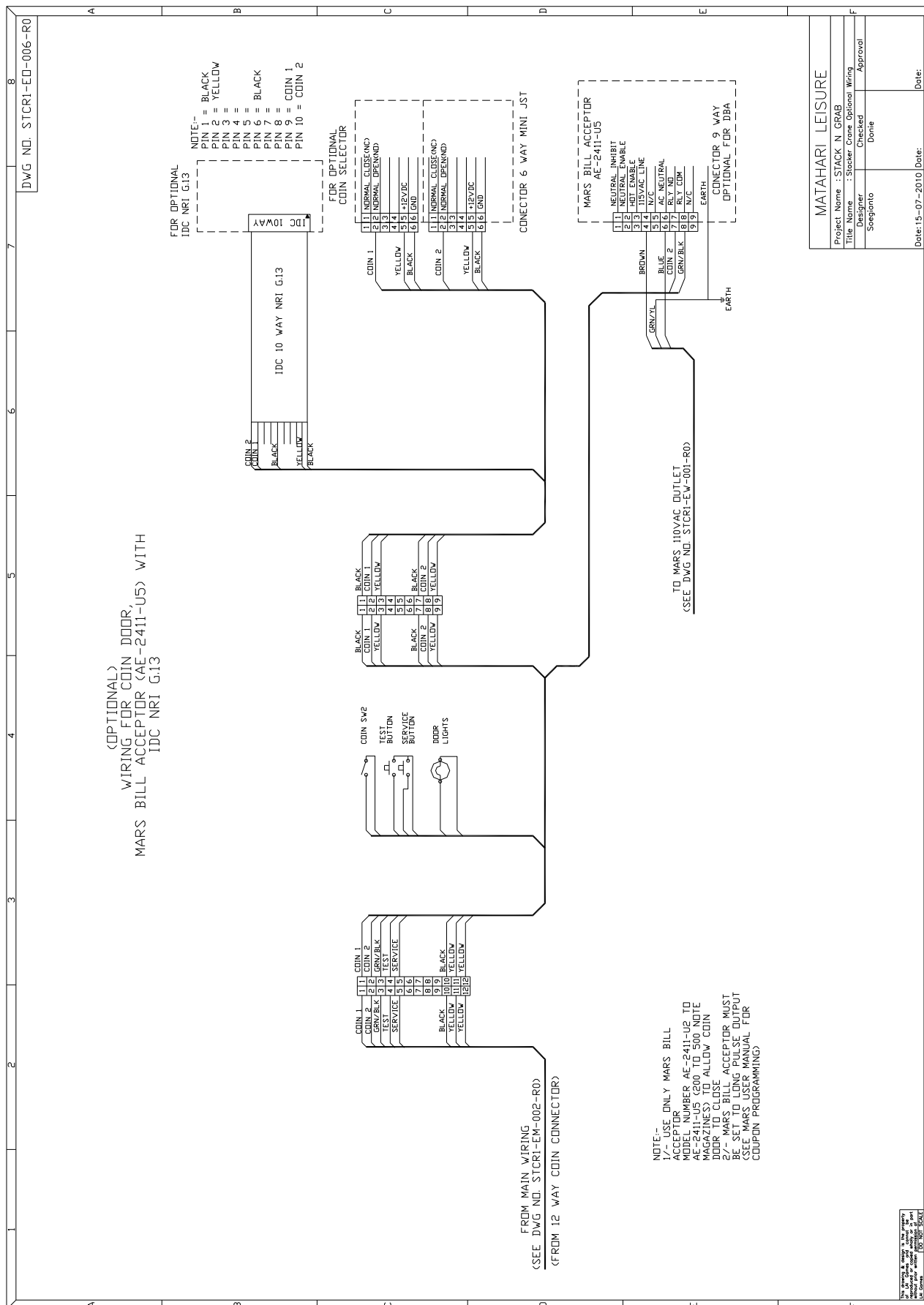
STACK 'N GRAB CONTROL WIRING





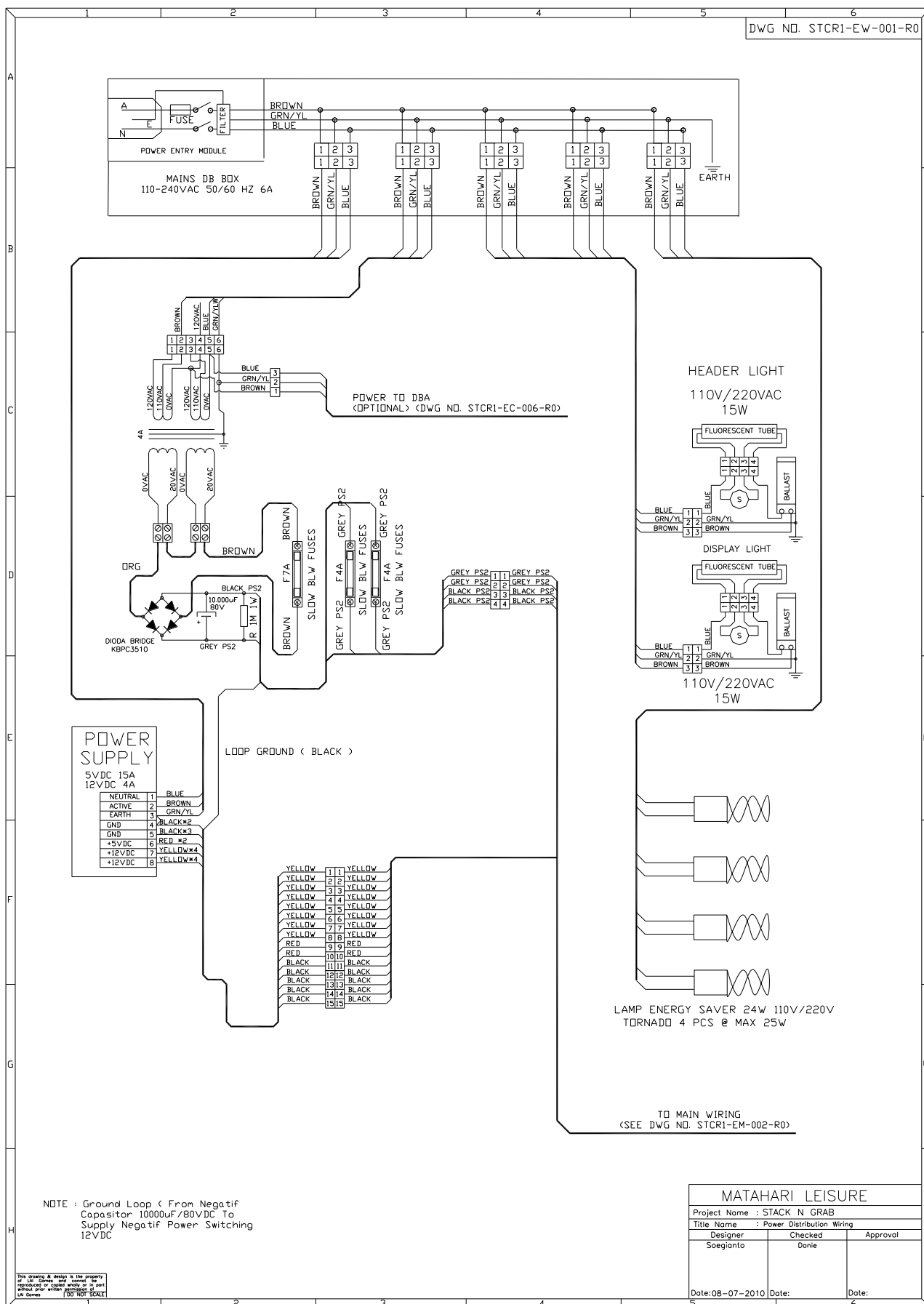


STACK 'N GRAB OPTIONAL WIRING





STACK N GRAB POWER WIRING



DISCLAIMER

OPERATOR WILL TAKE NOTE.

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Operator assumes any and all risk and liability for any civil or criminal legal claims or causes of action arising from the unauthorized use and/or operation of the provided hardware and licensed software, such improper and unauthorized use specifically including, but not limited to:

- (a) Operating or allowing the operation of the hardware and licensed software in a manner that violates the laws and regulations of the country, state, and/or municipality in which the hardware and licensed software are used or operated;
- (b) Assembling or causing the assembly of the hardware in a manner not authorized by or disclosed in this manual;
- (c) Any tampering with, changes to, or modifications of the licensed software that occur after the software leaves **LAI GAMES'** factory that is not made by authorized **LAI GAMES** personnel and that is directly or indirectly caused by Operator; and
- (d) Any tampering with the computer chip/electronic programmable read only memory (EPROM) by or on behalf of Operator that directly or indirectly causes the tamper-indicating holographic seal on the computer chip/EPROM to be broken or damaged in any way.

LAI GAMES shall have no liability related to such improper and unauthorized use and/or operation of the hardware and licensed software, and Operator shall indemnify, defend, and hold **LAI GAMES** harmless for any claim or cause of action brought against **LAI GAMES** arising from Operator's or Operator's representative's improper and unauthorized use and/or operation of the hardware and licensed software.

ANY IMPROPER AND UNAUTHORIZED USE SHALL COMPLETELY AND TOTALLY VOID ANY AND ALL WARRANTIES, BOTH EXPRESS AND IMPLIED, OF THE HARDWARE AND LICENSED SOFTWARE PROVIDED BY **LAI GAMES**.

WARRANTY

LAI Games warrants its manufactured products for a period of 3 months inclusive of parts and labor from the date of sale.

LAI Games exclusive obligation is to repair any item with any defects as a result of faulty workmanship or materials, providing the defective item or items of equipment are returned to the **LAI Games** distributor from which the machine was purchased.

LAI Games shall have no obligation to make repairs necessitated by negligence or interference to any component by any unauthorized personal. This will automatically void any existing warranty.

IF MAKING A WARRANTY CLAIM:

- (a) A Copy of the sales invoice must accompany the claim.
- (b) To and from Transport and freight costs are not covered by the warranty.
- (c) Warranty is not transferable with the sale of a machine from one owner to another.

